Laboratory Manual

Chemistry of the Functional Derivatives of Organic Molecules

Year 3

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Laboratory work No. 6–7

Carboxylic Acids. Preparation of m-Nitrobenzoic Acid

Topic: Carboxylic acids: structure, acidity and electrophilic substitution in aromatic acids. Preparation of m-nitrobenzoic acid

1. Relevance of the Topic

Carboxylic acids are among the most important classes of organic compounds in pharmacy. The carboxyl group is present in non-steroidal anti-inflammatory drugs (acetylsalicylic acid, ibuprofen, diclofenac), in amino acids, bile acids and many metabolites. Aromatic nitro acids are key intermediates in drug synthesis: nitrobenzoic acids are reduced to aminobenzoic acids, the building blocks of local anaesthetics and other drug substances.

The nitration of benzoic acid is a model reaction that shows how a substituent already present on the ring controls both the rate and the orientation of electrophilic aromatic substitution. The carboxyl group deactivates the ring and directs the incoming nitro group to the meta position. The same principle is used to plan the synthesis of substituted aromatic drug substances.

2. Learning Objectives

By the end of the session, the student must know:

- The electronic structure of the carboxyl group and the reasons for the acidity of carboxylic acids (resonance stabilisation of the carboxylate anion).
- The influence of substituents on acidity: electron-withdrawing groups increase it and electron-donating groups decrease it (pKa: benzoic acid 4.20, m-nitrobenzoic acid 3.45, o-nitrobenzoic acid 2.17).
- The mechanism of electrophilic aromatic nitration: generation of the nitronium ion, formation of the σ-complex, loss of a proton.
- Why the carboxyl group is a deactivating meta-director.

By the end of the session, the student must be able to:

- Carry out nitration with a nitrating mixture under strict temperature control.
- Isolate the product by pouring onto ice, vacuum filtration and recrystallisation.
- Confirm the acidic properties and the identity of the product by qualitative tests and melting point.
- Calculate the theoretical and percent yield.

3. Theoretical Background

3.1 Structure and acidity of carboxylic acids

In the carboxyl group –COOH, the carbonyl and hydroxyl groups are conjugated: the lone pair of the hydroxyl oxygen interacts with the π-bond of C=O (p,π-conjugation). This makes the O–H bond more polar than in alcohols and, above all, stabilises the carboxylate anion formed on ionisation. Its negative charge is delocalised equally over both oxygen atoms, and both C–O bonds become equal in length (about 0.127 nm).

As a result, carboxylic acids (pKa ≈ 4–5) are much stronger acids than alcohols (pKa ≈ 16) and phenols (pKa ≈ 10), but weaker than mineral acids. Carboxylic acids react with bases, active metals and — unlike phenols and alcohols — with sodium hydrogen carbonate, releasing CO2. This is a simple test that distinguishes carboxylic acids from phenols.

Substituents change acidity through inductive and resonance effects. Electron-withdrawing groups (–NO2, –Cl, –CN) stabilise the carboxylate anion and increase acidity; electron-donating groups (–CH3, –OCH3, –NH2) have the opposite effect. The ortho-nitro isomer is the strongest acid because the effect acts at the shortest distance (the ortho effect).

3.2 Electrophilic substitution in benzoic acid

The carboxyl group withdraws electron density from the ring by both inductive (−I) and resonance (−M) effects. Therefore benzoic acid undergoes electrophilic aromatic substitution (SEAr) more slowly than benzene (deactivation), and substitution occurs mainly at the meta position. The electrophile in nitration is the nitronium ion NO2+, generated from nitric acid by concentrated sulfuric acid:

$$HNO3 + 2H2SO4 \rightarrow NO2+ + H3O+ + 2HSO4-$$

The nitronium ion attacks the π-system of the ring to form a positively charged σ-complex (arenium ion), which then loses a proton to restore aromaticity. For attack at the ortho and para positions, one resonance structure of the σ-complex places the positive charge on the carbon bearing the electron-withdrawing –COOH group, which is strongly destabilising. Attack at the meta position avoids this, so the meta product predominates: nitration of benzoic acid gives about 80% meta, 19% ortho and 1–2% para isomer.

$$C6H5COOH + HNO3 \rightarrow m\text{-}O2N\text{–}C6H4\text{–}COOH + H2O \quad (H2SO4)$$

Because the ring is deactivated, a second nitration is much slower. Keeping the temperature low suppresses both dinitration and oxidative side reactions.

4. Self-Control Questions (Pre-lab Preparation)

1. Why are carboxylic acids stronger acids than alcohols and phenols? Illustrate your answer with the resonance structures of the carboxylate anion.
2. Arrange benzoic, m-nitrobenzoic, o-nitrobenzoic and p-methylbenzoic acids in order of increasing acidity and explain the order.
3. Write the equation for the formation of the nitronium ion. What is the role of concentrated sulfuric acid in the nitrating mixture?
4. Draw the σ-complexes for ortho, meta and para attack of NO2+ on benzoic acid and explain why meta substitution predominates.
5. Why must the temperature be kept low during nitration?
6. Why is the reaction mixture poured onto ice at the end of the reaction?
7. How can you show experimentally that the product is a carboxylic acid and not a phenol?
8. How is the purity of the product checked? What would a low and broad melting range indicate?
9. Which products are obtained by nitration of toluene, and why do they differ from the product of nitration of benzoic acid?

5. Reagents, Materials, and Equipment

Starting material: benzoic acid.

Reagent	Composition / notes	Used in
Concentrated sulfuric acid	H2SO4, 96–98% (d 1.84)	Exp. 1
Concentrated nitric acid	HNO3, 65–68% (d 1.40)	Exp. 1
Crushed ice, distilled water	quenching, washing, recrystallisation	Exp. 1
Sodium hydrogen carbonate solution	5–10% aqueous NaHCO3	Exp. 2
Sodium hydroxide solution	10% aqueous NaOH	Exp. 2
Hydrochloric acid	10% aqueous HCl	Exp. 2
Iron(III) chloride solution	3% aqueous FeCl3	Exp. 2

Equipment: conical flask (50 mL), thermometer (0–100 °C), dropping pipette, ice–salt bath, glass rod, Büchner funnel with filter flask, beakers for recrystallisation, test tubes, melting-point apparatus with capillaries, fume hood, safety goggles, gloves.

6. Laboratory Experiments

Experiment 1 — Nitration of Benzoic Acid (Synthesis of m-Nitrobenzoic Acid)

Principle: electrophilic aromatic substitution by the nitronium ion; the deactivating –COOH group directs the nitro group mainly to the meta position.

Procedure:

1. In the fume hood, place 2.0 g (16.4 mmol) of benzoic acid in a 50 mL conical flask and add 5 mL of concentrated H2SO4. Stir with a glass rod and cool the flask in an ice–salt bath to 0–5 °C.
2. In a separate test tube, prepare the nitrating mixture: slowly add 1.5 mL of concentrated HNO3 to 1.5 mL of concentrated H2SO4 with cooling. Cool the mixture in the ice bath.
3. Add the cold nitrating mixture dropwise to the benzoic acid solution with constant stirring, watching the thermometer: the temperature must not rise above 5–10 °C. The addition takes 10–15 minutes.
4. Keep the mixture in the ice bath with occasional stirring for another 10–15 minutes, then let it stand at room temperature for 10 minutes.
5. Pour the reaction mixture slowly, with stirring, onto 30–40 g of crushed ice in a beaker. The crude product separates as a white to pale-yellow precipitate.
6. Collect the solid by vacuum filtration and wash it on the filter with several small portions of ice-cold water until the washings are only weakly acidic.
7. Recrystallise the crude product from the minimum volume of boiling water, cool in ice, filter and dry the crystals in air. Weigh the product and calculate the yield.

$$C6H5COOH + HNO3 \rightarrow m\text{-}O2N\text{–}C6H4\text{–}COOH + H2O \quad (H2SO4,\ 0\text{–}10\ °C)$$

Result: white or pale-yellow crystals of m-nitrobenzoic acid (M = 167.12 g/mol), lit. m.p. 140–141 °C. Theoretical yield from 2.0 g of benzoic acid: 2.74 g.

Note: the ortho isomer formed as a by-product is more soluble in water than the meta isomer and remains mostly in the mother liquor after recrystallisation.

Experiment 2 — Acidic Properties of the Product

Principle: the carboxyl group is acidic enough to decompose hydrogen carbonate (release of CO2) and to form water-soluble salts; the free acid is regenerated by a stronger acid. Aromatic carboxylate anions form a coloured basic iron(III) salt in neutral solution.

Procedure:

1. Reaction with NaHCO3: to a few crystals of m-nitrobenzoic acid (and, in a parallel tube, of benzoic acid) add 1–2 mL of NaHCO3 solution and observe.
2. Salt formation and regeneration: dissolve about 0.1 g of the product in 1–2 mL of 10% NaOH, then add 10% HCl dropwise until the solution is acidic.
3. Reaction with FeCl3: dissolve a small amount of the acid in a few drops of NaOH, neutralise carefully with HCl to a neutral reaction (check with indicator paper — excess acid or base spoils the test), and add 2–3 drops of FeCl3 solution.

$$m\text{-}O2N\text{–}C6H4\text{–}COOH + NaHCO3 \rightarrow m\text{-}O2N\text{–}C6H4\text{–}COONa + CO2\uparrow + H2O$$

$$m\text{-}O2N\text{–}C6H4\text{–}COONa + HCl \rightarrow m\text{-}O2N\text{–}C6H4\text{–}COOH\downarrow + NaCl$$

Result: effervescence (CO2) with NaHCO3; the acid dissolves in NaOH and precipitates again on acidification; with FeCl3 a pale brown (buff) precipitate of the basic iron(III) salt forms.

Conclusion: the product is a carboxylic acid; a phenol would dissolve in NaOH but would not release CO2 from NaHCO3.

Experiment 3 — Melting Point Determination

Principle: a pure compound melts over a narrow range close to the literature value; impurities (the ortho isomer, unreacted benzoic acid) lower and broaden the melting range.

Procedure:

1. Dry a small sample of the recrystallised product, pack it 2–3 mm high into a capillary tube and determine the melting point, heating slowly (1–2 °C/min) near the expected value.
2. If time allows, determine the melting point of the starting benzoic acid (lit. 122 °C) for comparison.

Result: a sharp melting point at 140–141 °C confirms pure m-nitrobenzoic acid; a range starting well below 140 °C indicates contamination and the need for a second recrystallisation.

7. Safety Precautions

- The nitrating mixture (concentrated HNO3 + H2SO4) is extremely corrosive and a strong oxidiser. Work only in the fume hood and always add nitric acid to sulfuric acid slowly, with cooling.
- Nitration is strongly exothermic: add the nitrating mixture dropwise and keep the temperature below 10 °C. Overheating leads to side reactions and the release of toxic brown nitrogen dioxide fumes.
- Pour the acid reaction mixture onto ice — never add water to the concentrated acid mixture.
- Nitro compounds are toxic and can be absorbed through the skin; avoid contact with the product and wash your hands after work.
- Wear a lab coat, safety goggles, and gloves at all times; no food or drink in the laboratory.
- Know the location of the eyewash station, safety shower, and fire extinguisher before starting work.
- Collect acidic filtrates and washings in the designated acid waste container.

8. Individual Assignment and Report Requirements

Individual assignment: calculate the theoretical yield of m-nitrobenzoic acid from 2.0 g of benzoic acid and the percent yield from the mass you isolated. Write the mechanism of nitration of benzoic acid with all resonance structures of the meta σ-complex, and predict the main product of nitration of (a) methyl benzoate, (b) acetanilide, (c) chlorobenzene.

Protocol (laboratory report) requirements. For each experiment performed, the notebook must contain:

1. Name of the experiment and the reagents used.
2. Balanced equation(s) of the reactions.
3. Observations (colour, precipitate, gas, odour, mass, melting or boiling point) described precisely, not just “positive/negative”.
4. Calculations where required (theoretical and percent yield, specific rotation, etc.).
5. A short conclusion for each experiment linking the result to the structure of the compound.

Record your results in the table below and draw a conclusion about the identity and purity of the product.

Compound	Appearance	Mass (g)	Yield (%)	m.p. observed (°C)	m.p. lit. (°C)	NaHCO3 test
Benzoic acid (starting material)		2.0	—		122	
m-Nitrobenzoic acid (product)					140–141	

9. Recommended Literature

1. Clayden, J., Greeves, N., Warren, S. Organic Chemistry. Oxford University Press.
2. Morrison, R. T., Boyd, R. N. Organic Chemistry.
3. Furniss, B. S. et al. (Vogel). Vogel's Textbook of Practical Organic Chemistry.
4. Ahluwalia, V. K., Aggarwal, R. Comprehensive Practical Organic Chemistry: Preparations and Qualitative Analysis.
5. The official study guide and lecture notes for Chemistry of the Functional Derivatives of Organic Molecules (current program, Year 3).

Students should also consult the current national/institutional pharmacopoeia for official identification reactions and assay methods related to this topic.

Laboratory work No. 8

Reactions of Carboxylic Acids and Their Derivatives with Nucleophilic Reagents. Synthesis of n-Butyl Acetate and Isobutyl Acetate

Topic: Nucleophilic acyl substitution. Esterification (Fischer reaction). Synthesis of n-butyl acetate and isobutyl acetate

1. Relevance of the Topic

Nucleophilic substitution at the acyl carbon is the central reaction type of carboxylic acids and their functional derivatives — acyl halides, anhydrides, esters and amides. It underlies both the synthesis and the breakdown of many drug substances: acetylsalicylic acid and paracetamol are made by acylation, local anaesthetics such as procaine and benzocaine are esters, and the hydrolysis of ester and amide bonds determines the stability and shelf life of these drugs as well as their metabolism in the body (esterases, amidases).

Esters of lower acids and alcohols are used as solvents, extractants and flavouring agents in pharmaceutical technology. The acid-catalysed synthesis of butyl acetates shows how an equilibrium reaction can be driven towards the product, and trains basic preparative techniques: reflux, washing in a separating funnel, drying and distillation.

2. Learning Objectives

By the end of the session, the student must know:

- The mechanism of nucleophilic acyl substitution (addition–elimination through a tetrahedral intermediate).
- The relative reactivity of carboxylic acid derivatives (acyl halides > anhydrides > esters ≈ acids > amides) and its explanation.
- The mechanism of acid-catalysed esterification (Fischer reaction) and the ways of shifting its equilibrium.
- Reactions of esters with nucleophiles: acid and alkaline hydrolysis (saponification), aminolysis, reaction with hydroxylamine.

By the end of the session, the student must be able to:

- Carry out an esterification under reflux and work up the reaction mixture (washing, neutralisation, drying).
- Purify a liquid product by distillation and identify it by boiling point (and refractive index, if a refractometer is available).
- Detect the ester group by the hydroxamic acid (iron(III) hydroxamate) test and by saponification.
- Calculate the theoretical and percent yield using the limiting reagent.

3. Theoretical Background

3.1 Nucleophilic acyl substitution

Carboxylic acids and their derivatives R–CO–X (X = Cl, OCOR, OR′, NH2, OH) react with nucleophiles by a two-step addition–elimination mechanism. The nucleophile attacks the electrophilic carbonyl carbon and forms a tetrahedral intermediate; the intermediate then collapses, the C=O bond is restored and the group X leaves. The overall result is the replacement of X by the nucleophile, while the acyl group R–CO– is preserved.

The reactivity of derivatives decreases in the series: acyl halides > anhydrides > esters ≈ carboxylic acids > amides > carboxylate anions. Two factors determine it: the ability of X to leave (Cl− is the best leaving group, NH2− the poorest), and the extent to which the lone pair of X donates electron density to the carbonyl carbon by resonance, reducing its electrophilicity (strong in amides, negligible in acyl chlorides). A more reactive derivative can always be converted into a less reactive one, but not directly the other way round.

Typical reactions with nucleophiles: hydrolysis (H2O or OH−) gives acids or carboxylates; alcoholysis (R′OH) gives esters; aminolysis (NH3, amines) gives amides; hydroxylamine gives hydroxamic acids; SOCl2 or PCl5 convert the OH group of an acid into Cl, giving the most reactive acyl chlorides.

3.2 Esterification (Fischer reaction)

Carboxylic acids react with alcohols only in the presence of a strong acid catalyst (concentrated H2SO4, p-toluenesulfonic acid). The catalyst protonates the carbonyl oxygen and makes the carbonyl carbon electrophilic enough to be attacked by the weakly nucleophilic alcohol. Proton transfer, loss of water and deprotonation follow. Isotope labelling (18O in the alcohol) shows that the ester oxygen comes from the alcohol: the C–OH bond of the acid is broken, not the O–H bond.

$$CH3COOH + CH3CH2CH2CH2OH \rightleftharpoons CH3COOCH2CH2CH2CH3 + H2O \quad (H2SO4)$$

$$CH3COOH + (CH3)2CHCH2OH \rightleftharpoons CH3COOCH2CH(CH3)2 + H2O \quad (H2SO4)$$

Esterification is reversible (equilibrium constant K ≈ 4 for primary alcohols), so with equimolar amounts only about two thirds of the reactants are converted. The yield is increased by using an excess of the cheaper reagent (here acetic acid) or by removing water (azeotropic distillation with a Dean–Stark trap, dehydrating agents). Acid hydrolysis of esters is the reverse reaction and is also catalysed by acids; alkaline hydrolysis (saponification) is irreversible because the carboxylate anion formed is no longer attacked by the alcohol.

4. Self-Control Questions (Pre-lab Preparation)

1. Write the general mechanism of nucleophilic acyl substitution and explain why it proceeds through a tetrahedral intermediate.
2. Arrange acetyl chloride, acetic anhydride, ethyl acetate and acetamide in order of decreasing reactivity towards water and explain the order.
3. Why does esterification require an acid catalyst? Write the mechanism of the formation of n-butyl acetate.
4. How can the equilibrium of esterification be shifted towards the ester? Which reagent is used in excess in this work, and why?
5. Why is alkaline hydrolysis of esters irreversible, while acid hydrolysis is reversible?

6. What is the purpose of washing the crude ester with water and with NaHCO3 solution? Why must the separating funnel be vented frequently?
7. What type of isomerism relates n-butyl acetate and isobutyl acetate? Which of them has the higher boiling point, and why?
8. Write the equations of the hydroxamic acid test for esters.
9. Give examples of drug substances that are esters or amides and explain why their hydrolysis matters for drug stability.

5. Reagents, Materials, and Equipment

Starting materials: n-butanol (butan-1-ol), isobutanol (2-methylpropan-1-ol), glacial acetic acid.

Reagent	Composition / notes	Used in
Concentrated sulfuric acid	H2SO4, 96–98% — catalyst	Exp. 1, 2
Sodium hydrogen carbonate solution	5% aqueous NaHCO3 — neutralisation of acids	Exp. 1, 2
Saturated sodium chloride solution	brine — reduces solubility of the ester in water	Exp. 1, 2
Anhydrous sodium sulfate (or magnesium sulfate)	drying agent	Exp. 1, 2
Hydroxylamine hydrochloride solution	1 M NH2OH·HCl in water or ethanol	Exp. 3
Sodium hydroxide solutions	10% and 2 M aqueous NaOH	Exp. 3, 4
Hydrochloric acid	1 M aqueous HCl	Exp. 3
Iron(III) chloride solution	5% aqueous FeCl3	Exp. 3

Equipment: round-bottom flask (50 mL) with reflux condenser, heating mantle or water/sand bath, boiling chips, separating funnel (100 mL), conical flasks, simple distillation apparatus (distillation flask, thermometer, Liebig condenser, receiver), graduated cylinders, test tubes, refractometer (optional), fume hood, safety goggles and gloves.

6. Laboratory Experiments

Experiment 1 — Synthesis of n-Butyl Acetate

Principle: acid-catalysed esterification of acetic acid with n-butanol; a twofold excess of acetic acid shifts the equilibrium towards the ester.

Procedure:

1. Place 9.2 mL (7.4 g, 0.10 mol) of n-butanol and 11.5 mL (12.0 g, 0.20 mol) of glacial acetic acid in a 50 mL round-bottom flask. Carefully add 0.5 mL of concentrated H2SO4 with swirling, then 2–3 boiling chips.
2. Attach a reflux condenser and heat the mixture at gentle boiling for 45–60 minutes.
3. Cool the flask to room temperature and pour the mixture into a separating funnel containing 20 mL of cold water. Shake, let the layers separate and discard the lower aqueous layer (the ester, d ≈ 0.88 g/mL, is the upper layer).
4. Wash the organic layer with 10–15 mL portions of 5% NaHCO3 until no more CO2 is evolved (vent the funnel frequently!), then with 10 mL of saturated NaCl solution.

5. Transfer the ester to a dry conical flask and dry it over 1–2 g of anhydrous Na2SO4 for 10–15 minutes, swirling occasionally, until the liquid is clear.
6. Decant the dried liquid into a dry distillation flask, add a boiling chip and distil, collecting the fraction boiling at 122–127 °C in a pre-weighed receiver. Weigh the product and calculate the yield.

$$CH3COOH + C4H9OH \rightleftharpoons CH3COOC4H9 + H2O \quad (H2SO4,\ t)$$

Result: a colourless liquid with a pleasant fruity odour; b.p. 126 °C, d20 = 0.882 g/mL, nD20 = 1.3941. Theoretical yield (limiting reagent n-butanol): 11.6 g.

Experiment 2 — Synthesis of Isobutyl Acetate

Principle: the same esterification with the branched isomeric alcohol. The product is a structural isomer of n-butyl acetate with a lower boiling point because its branched molecules are more compact and have weaker intermolecular (dispersion) forces.

Procedure:

1. Carry out the synthesis exactly as in Experiment 1, using 9.3 mL (7.4 g, 0.10 mol) of isobutanol instead of n-butanol.
2. In the distillation step, collect the fraction boiling at 114–119 °C.

$$CH3COOH + (CH3)2CHCH2OH \rightleftharpoons CH3COOCH2CH(CH3)2 + H2O \quad (H2SO4,\ t)$$

Result: a colourless liquid with a fruity odour; b.p. 117–118 °C, d20 = 0.871 g/mL, nD20 = 1.3902. Theoretical yield: 11.6 g.

Note: the work is usually divided so that half of the group prepares n-butyl acetate and half isobutyl acetate; the results are compared in the summary table.

Experiment 3 — Hydroxamic Acid Test for the Ester Group

Principle: esters react with hydroxylamine in alkaline medium (nucleophilic acyl substitution) to form hydroxamic acids, which give intensely coloured red-violet complexes with Fe3+ ions.

Procedure:

1. To 2–3 drops of the ester in a test tube add 1 mL of hydroxylamine hydrochloride solution and 0.5 mL of 10% NaOH.
2. Heat the mixture to boiling for 1–2 minutes and cool.
3. Acidify with 1 M HCl (check with indicator paper) and add 1–3 drops of FeCl3 solution.
4. Run a blank test with n-butanol for comparison.

$$CH3COOC4H9 + NH2OH \rightarrow CH3CO\text{–}NHOH + C4H9OH$$

$$3\ CH3CO\text{–}NHOH + Fe3+ \rightarrow Fe(CH3CO\text{–}NHO)3 + 3H+$$

Result: a red-violet (burgundy) colour with the ester; the alcohol gives no colour other than the yellow of FeCl3.

Experiment 4 — Alkaline Hydrolysis (Saponification) of the Ester

Principle: the hydroxide ion is a strong nucleophile; it converts the ester irreversibly into the carboxylate salt and the alcohol.

Procedure:

1. Place 0.5 mL of the ester and 5 mL of 2 M NaOH in a test tube; note the two layers and the fruity odour.

2. Heat the tube in a boiling water bath for 10–15 minutes, shaking frequently.
3. Observe the gradual disappearance of the ester layer and the change of odour.

$$CH3COOC4H9 + NaOH \rightarrow CH3COONa + C4H9OH$$

Result: the upper ester layer gradually disappears and the fruity odour is replaced by the odour of butanol; the solution becomes homogeneous.

7. Safety Precautions

- Concentrated sulfuric acid and glacial acetic acid are corrosive; acetic acid vapour is irritating. Measure them in the fume hood and add H2SO4 slowly with swirling.
- Butanols and butyl acetates are flammable liquids with irritating vapours. Heat only with a heating mantle or water/sand bath — never with an open flame.
- Always add boiling chips before heating; never add them to a hot liquid.
- When washing with NaHCO3, CO2 is released: invert the separating funnel and open the tap frequently to release pressure, pointing the outlet away from people.
- Never distil to dryness; stop the distillation while a small residue remains in the flask.
- Hydroxylamine hydrochloride is harmful and a skin sensitiser — avoid contact.
- Wear a lab coat, safety goggles, and gloves at all times; no food or drink in the laboratory.
- Know the location of the eyewash station, safety shower, and fire extinguisher before starting work.
- Collect organic residues in the designated organic waste container.

8. Individual Assignment and Report Requirements

Individual assignment: calculate the theoretical and percent yield of your ester. Using the equilibrium constant K = 4, calculate the maximum conversion of the alcohol when (a) equimolar amounts and (b) a twofold excess of acetic acid are used, and compare the result with your experimental yield.

Protocol (laboratory report) requirements. For each experiment performed, the notebook must contain:

1. Name of the experiment and the reagents used.
2. Balanced equation(s) of the reactions.
3. Observations (colour, precipitate, gas, odour, mass, melting or boiling point) described precisely, not just “positive/negative”.
4. Calculations where required (theoretical and percent yield, specific rotation, etc.).
5. A short conclusion for each experiment linking the result to the structure of the compound.

Record the results for both esters in the table below and compare the boiling points and yields of the two isomers.

Ester	Alcohol used	b.p. fraction (°C)	b.p. lit. (°C)	Mass (g)	Yield (%)	Hydroxamic test
n-Butyl acetate	n-Butanol		126			
Isobutyl acetate	Isobutanol		117–118			

9. Recommended Literature

1. Clayden, J., Greeves, N., Warren, S. Organic Chemistry. Oxford University Press.
2. Morrison, R. T., Boyd, R. N. Organic Chemistry.
3. Furniss, B. S. et al. (Vogel). Vogel's Textbook of Practical Organic Chemistry.
4. Ahluwalia, V. K., Aggarwal, R. Comprehensive Practical Organic Chemistry: Preparations and Qualitative Analysis.
5. The official study guide and lecture notes for Chemistry of the Functional Derivatives of Organic Molecules (current program, Year 3).

Students should also consult the current national/institutional pharmacopoeia for official identification reactions and assay methods related to this topic.

Laboratory work No. 9

Diazo and Azo Compounds. Synthesis of Iodobenzene

Topic: Diazotization of primary aromatic amines. Reactions of arenediazonium salts with and without loss of nitrogen. Synthesis of iodobenzene

1. Relevance of the Topic

Arenediazonium salts are among the most versatile intermediates in aromatic chemistry. The diazo group can be replaced by almost any substituent (–OH, –F, –Cl, –Br, –I, –CN, –H), which gives substitution patterns that cannot be obtained by direct electrophilic substitution. Aryl iodides such as iodobenzene are important starting materials for modern cross-coupling reactions used in drug synthesis.

Diazotization is also a key analytical reaction in pharmacy. The pharmacopoeial identification test for a primary aromatic amino group (sulfonamides, procaine, benzocaine) is based on diazotization followed by azo coupling with 2-naphthol, and the quantitative nitritometric (sodium nitrite) titration is based on the same reaction.

2. Learning Objectives

By the end of the session, the student must know:

- The mechanism of diazotization and the conditions required (excess acid, 0–5 °C).
- The structure and relative stability of arenediazonium salts, and why they must be kept cold and used in solution.
- Reactions of diazonium salts with loss of nitrogen (hydrolysis, replacement by halogens — Sandmeyer and related reactions, the Schiemann reaction, reduction) and without loss of nitrogen (azo coupling).
- Why iodide replaces the diazo group without a copper catalyst.

By the end of the session, the student must be able to:

- Carry out diazotization of aniline with temperature control and detect the end point with starch–iodide paper.
- Replace the diazo group by iodine, and isolate and purify the liquid product.
- Perform the reactions of the diazonium salt with water (on warming) and with 2-naphthol (azo coupling).
- Calculate the yield.

3. Theoretical Background

3.1 Diazotization

Primary aromatic amines react with nitrous acid (generated in situ from sodium nitrite and hydrochloric acid) at 0–5 °C to form arenediazonium salts:

$$C6H5NH2 + NaNO2 + 2HCl \rightarrow C6H5N2+ Cl- + NaCl + 2H2O$$

The electrophile is the nitrosonium ion NO+ (or its carriers N2O3, NOCl), formed from nitrous acid in acidic medium. It attacks the nitrogen of the free amine to form an N-nitrosamine, which isomerises to a

diazohydroxide; protonation and loss of water give the diazonium ion. An excess of acid (2.5–3 equivalents) is needed to generate HNO2, to keep the amine in solution as its salt, and to prevent coupling of the diazonium salt with unreacted amine.

Arenediazonium ions are stabilised by conjugation of the –N≡N+ group with the aromatic ring, which is why, unlike alkyldiazonium ions, they can exist in cold aqueous solution. On warming they decompose with evolution of nitrogen, and in the dry state many diazonium salts explode on impact or heating. Diazonium salts are therefore always prepared at 0–5 °C and used immediately in solution.

3.2 Reactions of arenediazonium salts

Reactions with loss of nitrogen. N2 is an excellent leaving group, so the diazo group is easily replaced: by OH on warming in water (phenols); by Cl, Br or CN with copper(I) salts (Sandmeyer reaction); by F on heating the tetrafluoroborate (Schiemann reaction); by H with hypophosphorous acid (deamination); and by I with potassium iodide. Iodide needs no copper catalyst: it is easily oxidised and transfers an electron to the diazonium ion, generating an aryl radical that combines with iodine.

$$C6H5N2+\ Cl- + KI \rightarrow C6H5I + N2\uparrow + KCl$$

$$C6H5N2+\ Cl- + H2O \rightarrow C6H5OH + N2\uparrow + HCl \quad (t)$$

Reactions without loss of nitrogen. The diazonium ion is a weak electrophile and couples with strongly activated aromatic compounds — phenols (in weakly alkaline medium) and aromatic amines (in weakly acidic medium) — to form coloured azo compounds Ar–N=N–Ar′ (azo dyes).

$$C6H5N2+\ Cl- + 2\text{-}naphthol + NaOH \rightarrow 1\text{-}phenylazo\text{-}2\text{-}naphthol\downarrow + NaCl + H2O$$

4. Self-Control Questions (Pre-lab Preparation)

1. Write the equation of diazotization of aniline and name the electrophilic species that attacks the amine.
2. Why is diazotization carried out at 0–5 °C and with an excess of hydrochloric acid?
3. How is the end of diazotization detected? Write the equation of the reaction on starch–iodide paper.
4. Why are arenediazonium salts more stable than alkyldiazonium salts? Why must they never be isolated in the dry state?
5. Classify the reactions of diazonium salts into reactions with and without loss of nitrogen and give an example of each.
6. Why does the replacement of the diazo group by iodine not require a copper catalyst, while replacement by chlorine or bromine does?
7. Why is the reaction mixture made alkaline before isolation of iodobenzene?
8. Why is azo coupling with 2-naphthol carried out in alkaline solution?
9. Propose syntheses of m-bromotoluene and 1,3,5-tribromobenzene using diazonium salts.
10. Which reaction is the basis of the pharmacopoeial test for a primary aromatic amino group?

5. Reagents, Materials, and Equipment

Starting material: aniline (freshly distilled, colourless or pale yellow).

Reagent	Composition / notes	Used in
Concentrated hydrochloric acid	HCl, 36%	Exp. 1
Sodium nitrite	NaNO2, solid; freshly prepared aqueous solution	Exp. 1
Starch–iodide paper	test for excess nitrous acid	Exp. 1
Potassium iodide	KI, solid	Exp. 2
Sodium hydroxide solution	10% aqueous NaOH	Exp. 2, 4
Sodium thiosulfate solution	5% aqueous Na2S2O3 — removes free iodine	Exp. 2
Anhydrous calcium chloride	drying agent	Exp. 2
Bromine water	saturated aqueous Br2	Exp. 3
2-Naphthol	solid	Exp. 4
Ice	cooling	Exp. 1–4

Equipment: conical flasks and beakers (100–250 mL), thermometer, ice–salt bath, dropping pipette, glass rods, water bath, separating funnel, steam distillation apparatus (if available), distillation apparatus with air condenser, test tubes, fume hood, safety goggles and gloves.

6. Laboratory Experiments

Experiment 1 — Diazotization of Aniline (Benzenediazonium Chloride Solution)

Principle: reaction of aniline hydrochloride with nitrous acid, generated from NaNO2 and HCl, at 0–5 °C; the end point is detected by a slight excess of HNO2, which oxidises iodide to iodine (blue colour with starch).

Procedure:

1. In the fume hood, place 6.5 mL of concentrated HCl and 6.5 mL of water in a 100 mL conical flask and add 2.3 mL (2.33 g, 25 mmol) of aniline with stirring. Aniline hydrochloride is formed (the mixture warms up).
2. Cool the solution in an ice–salt bath to 0–5 °C (some aniline hydrochloride may crystallise; this does not interfere).
3. Dissolve 1.8 g (26 mmol) of NaNO2 in 5 mL of water and cool the solution.
4. Add the nitrite solution dropwise with constant stirring, keeping the temperature below 5 °C (add small pieces of ice directly to the mixture if necessary).
5. Towards the end of the addition, test the mixture with starch–iodide paper 1–2 minutes after each portion. Diazotization is complete when a drop of the mixture turns the paper blue immediately.
6. Keep the clear diazonium solution in the ice bath and use it at once: withdraw two 1–2 mL portions for Experiments 3 and 4 and use the rest for Experiment 2.

$$C6H5NH2 + HCl \rightarrow C6H5NH3+ Cl-$$

$$C6H5NH3+ Cl- + NaNO2 + HCl \rightarrow C6H5N2+ Cl- + NaCl + 2H2O$$

$$2HNO2 + 2KI + 2HCl \rightarrow I2 + 2NO\uparrow + 2KCl + 2H2O \quad (starch\text{–}iodide\ test)$$

Result: a clear, almost colourless or pale-yellow solution of benzenediazonium chloride; the starch–iodide paper turns blue at the end point.

Experiment 2 — Synthesis of Iodobenzene

Principle: iodide ion replaces the diazo group with evolution of nitrogen; no catalyst is required.

Procedure:

1. Dissolve 4.5 g (27 mmol) of KI in 7 mL of water and add this solution slowly, with stirring, to the cold diazonium solution. A brown colour appears and nitrogen is evolved.
2. Leave the mixture at room temperature for 15–20 minutes, then heat it on a water bath (60–70 °C) until the evolution of nitrogen stops. A heavy dark oil (crude iodobenzene) collects at the bottom.
3. Cool the mixture and make it strongly alkaline with 10% NaOH; this converts the phenol by-product into water-soluble sodium phenoxide.
4. Isolate the iodobenzene by steam distillation. If steam distillation is not available, transfer the mixture to a separating funnel, run off the lower (heavy) organic layer and wash it with 5% Na2S2O3 solution (to remove iodine) and then with water.
5. Dry the crude product over anhydrous CaCl2, then distil it using an air condenser, collecting the fraction boiling at 185–190 °C in a pre-weighed receiver. Weigh the product and calculate the yield.

$$C6H5N2+ Cl- + KI \rightarrow C6H5I + N2\uparrow + KCl$$

Result: a colourless or pale-yellow heavy liquid, b.p. 188 °C, d = 1.83 g/mL, which darkens on exposure to light (liberation of iodine). Theoretical yield from 25 mmol of aniline: 5.1 g.

Note: store iodobenzene in a dark bottle; if it turns brown, wash it with thiosulfate solution before use.

Experiment 3 — Hydrolysis of the Diazonium Salt (Reaction with Loss of Nitrogen)

Principle: on warming, the diazonium ion reacts with water: nitrogen is released and phenol is formed.

Procedure:

1. Warm 1–2 mL of the diazonium solution in a water bath at 50–60 °C. Observe the evolution of gas and note the odour of phenol.
2. Cool the tube and add bromine water dropwise.

$$C6H5N2+ Cl- + H2O \rightarrow C6H5OH + N2\uparrow + HCl$$

$$C6H5OH + 3Br2 \rightarrow 2,4,6\text{-}Br3C6H2OH\downarrow + 3HBr$$

Result: gas bubbles (N2) and the characteristic odour of phenol; bromine water is decolourised and a white precipitate of 2,4,6-tribromophenol forms.

Experiment 4 — Azo Coupling with 2-Naphthol (Reaction without Loss of Nitrogen)

Principle: the diazonium ion couples with the 2-naphthoxide ion at position 1, forming the orange-red azo dye 1-phenylazo-2-naphthol.

Procedure:

1. Dissolve about 0.1 g of 2-naphthol in 2 mL of 10% NaOH and cool the solution in ice.
2. Add 1 mL of the cold diazonium solution with shaking.

$$C6H5N2+ Cl- + C10H7OH + NaOH \rightarrow C6H5\text{–}N{=}N\text{–}C10H6OH\downarrow + NaCl + H2O$$

Result: an orange-red precipitate of the azo dye forms immediately.

Practical significance: the same sequence — diazotization and coupling with 2-naphthol — is the pharmacopoeial identification test for drugs containing a primary aromatic amino group.

7. Safety Precautions

- Aniline is toxic by all routes of exposure and is readily absorbed through the skin (it causes methaemoglobinaemia); it is a suspected carcinogen and mutagen. Measure it in the fume hood and wear gloves.
- Diazonium salts are unstable: keep the solution at 0–5 °C, use it immediately, and never let it evaporate or dry — dry diazonium salts can explode. Destroy any unused diazonium solution by warming it in the fume hood until nitrogen evolution stops.
- Sodium nitrite is toxic and an oxidiser. Nitrogen oxides formed from excess nitrous acid are toxic — work in the fume hood.
- Concentrated HCl is corrosive; NaOH solutions are caustic.
- Iodobenzene is harmful and irritating. For distillation at 188 °C use an air condenser (a water-cooled condenser may crack), and never distil to dryness.
- 2-Naphthol and bromine water are harmful; bromine vapour is toxic.
- Wear a lab coat, safety goggles, and gloves at all times; no food or drink in the laboratory.
- Know the location of the eyewash station, safety shower, and fire extinguisher before starting work.

8. Individual Assignment and Report Requirements

Individual assignment: (1) calculate the theoretical and percent yield of iodobenzene. (2) Starting from aniline, propose syntheses of chlorobenzene, benzonitrile, fluorobenzene and phenol via the diazonium salt; name each reaction and state the conditions.

Protocol (laboratory report) requirements. For each experiment performed, the notebook must contain:

1. Name of the experiment and the reagents used.
2. Balanced equation(s) of the reactions.
3. Observations (colour, precipitate, gas, odour, mass, melting or boiling point) described precisely, not just “positive/negative”.
4. Calculations where required (theoretical and percent yield, specific rotation, etc.).
5. A short conclusion for each experiment linking the result to the structure of the compound.

Record your observations in the table below, including the mass, boiling point and yield of iodobenzene.

Reaction	Reagent / conditions	Observation	Product
Diazotization	NaNO2, HCl, 0–5 °C		
Replacement by iodine	KI, then warming		
Hydrolysis	H2O, 50–60 °C		
Azo coupling	2-naphthol, NaOH		
Iodobenzene (isolated)	b.p. ____ °C	mass ____ g	yield ____ %

9. Recommended Literature

1. Clayden, J., Greeves, N., Warren, S. Organic Chemistry. Oxford University Press.
2. Morrison, R. T., Boyd, R. N. Organic Chemistry.

3. Furniss, B. S. et al. (Vogel). Vogel's Textbook of Practical Organic Chemistry.
4. Ahluwalia, V. K., Aggarwal, R. Comprehensive Practical Organic Chemistry: Preparations and Qualitative Analysis.
5. The official study guide and lecture notes for Chemistry of the Functional Derivatives of Organic Molecules (current program, Year 3).

Students should also consult the current national/institutional pharmacopoeia for official identification reactions and assay methods related to this topic.

Laboratory work No. 10

Qualitative Reactions of Polycarboxylic Acids (Oxo- and Hydroxy Acids). Qualitative Reactions of Phenols

Topic: Dicarboxylic, hydroxy and oxo acids: mutual influence of functional groups. Phenols: acidity and reactivity. Qualitative reactions

1. Relevance of the Topic

Hydroxy and oxo acids — lactic, malic, tartaric, citric, pyruvic and acetoacetic acids — are key metabolites of carbohydrate and fat metabolism (glycolysis, the citric acid cycle, ketone body formation). Many of them are used in pharmacy as active substances, excipients and buffer components (citric and tartaric acids, calcium lactate, salicylic acid). Oxalic acid forms poorly soluble calcium oxalate, the main component of most kidney stones.

Phenols are an equally important group: phenol and its derivatives are antiseptics, and the phenolic hydroxyl group occurs in salicylic acid, paracetamol, adrenaline, morphine, estradiol and tocopherol. Qualitative reactions of these compounds are used in pharmacopoeial identification tests and in clinical laboratory diagnostics.

2. Learning Objectives

By the end of the session, the student must know:

- The classification of polyfunctional and heterofunctional carboxylic acids: dicarboxylic, hydroxy and oxo acids, phenolic acids.
- Properties arising from the mutual influence of functional groups: higher acidity of dicarboxylic and α-hydroxy acids, oxidation of oxalic acid, chelation of metal ions by hydroxy acids, keto–enol tautomerism of β-oxo acids and their esters.
- The acidity of phenols compared with alcohols and carboxylic acids, and the high reactivity of the phenol ring in electrophilic substitution.
- The chemistry of the qualitative reactions performed in this work.

By the end of the session, the student must be able to:

- Identify oxalic, lactic, tartaric and citric acids and ethyl acetoacetate by their characteristic reactions.
- Distinguish a phenol from an alcohol and from a carboxylic acid using FeCl3, bromine water, NaOH and NaHCO3.
- Explain every observation with a balanced equation or a structural argument.

3. Theoretical Background

3.1 Polycarboxylic, hydroxy and oxo acids

Dicarboxylic acids (oxalic HOOC–COOH, malonic, succinic) are stronger acids than monocarboxylic acids in their first dissociation step, because one carboxyl group withdraws electron density from the other (oxalic acid pKa1 = 1.25). Oxalic acid is easily oxidised to CO2 — it is used as a primary standard in permanganatometry — and forms a very poorly soluble calcium salt.

Hydroxy acids contain both hydroxyl and carboxyl groups: lactic acid CH3–CH(OH)–COOH, tartaric acid HOOC–CH(OH)–CH(OH)–COOH, citric acid HOOC–CH2–C(OH)(COOH)–CH2–COOH, and the phenolic salicylic acid. The neighbouring OH group increases acidity (−I effect) and allows hydroxy acids to form stable chelate complexes with metal ions: tartrate keeps Cu2+ in solution in Fehling's reagent, citrate in Benedict's reagent. Lactic acid, which contains the CH3–CH(OH)– fragment, gives the iodoform reaction.

Oxo acids contain a carbonyl group: pyruvic acid CH3–CO–COOH (an α-oxo acid) and acetoacetic acid CH3–CO–CH2–COOH (a β-oxo acid). β-Oxo acids decarboxylate easily; they and their esters (ethyl acetoacetate) exist as an equilibrium mixture of keto and enol tautomers. The enol form gives a colour reaction with FeCl3 and decolourises bromine water:

$$CH3–CO–CH2–COOC2H5\ (keto\ form) \rightleftharpoons CH3–C(OH)=CH–COOC2H5\ (enol\ form)$$

3.2 Phenols

In phenols the hydroxyl group is bonded directly to the aromatic ring. The lone pair of oxygen is conjugated with the π-system (+M effect). This (1) makes phenols more acidic than alcohols, because the phenoxide ion is stabilised by delocalisation of the negative charge into the ring (phenol pKa ≈ 10, ethanol ≈ 16), and (2) strongly activates the ring towards electrophilic substitution at the ortho and para positions.

Phenols are weaker acids than carboxylic acids and carbonic acid: they dissolve in NaOH but do not decompose NaHCO3. This difference is used to separate and distinguish phenols from carboxylic acids. Because of the activated ring, phenol reacts instantly with bromine water without a catalyst, giving a white precipitate of 2,4,6-tribromophenol. Phenols (and enols) form coloured complexes with Fe3+ ions; the colour depends on the structure: violet for phenol and salicylic acid, dark violet for resorcinol, green for pyrocatechol; hydroquinone is oxidised by Fe3+ to p-benzoquinone.

4. Self-Control Questions (Pre-lab Preparation)

1. Why is oxalic acid a stronger acid than acetic acid? Write the equation of its oxidation by KMnO4 in acidic medium.
2. Which reaction is used to detect oxalate ions? Why is it important in medicine?
3. Which structural fragment of lactic acid is responsible for the positive iodoform test?
4. Explain the principle of Uffelmann's test for lactic acid.
5. Why do tartaric and citric acids dissolve copper(II) hydroxide in alkaline medium? Where is this property used?
6. How can tartaric acid be distinguished from citric acid using calcium ions?
7. Write the keto–enol equilibrium of ethyl acetoacetate and explain the colour changes observed with FeCl3 and bromine water.
8. Explain why phenol is a stronger acid than ethanol but a weaker acid than acetic acid.
9. Write the equation of the reaction of phenol with bromine water. Why does it proceed without a catalyst?
10. How can phenol, benzyl alcohol and benzoic acid be distinguished using NaOH, NaHCO3 and FeCl3?

5. Reagents, Materials, and Equipment

Test substances: oxalic acid (5% solution), lactic acid (1–2% solution), tartaric acid (10% solution), citric acid (10% solution), ethyl acetoacetate, phenol (1% solution and phenol emulsion), resorcinol, hydroquinone and salicylic acid (1% solutions), ethanol.

Reagent	Composition / notes	Used in
Calcium chloride solution	5% aqueous CaCl2	Exp. 1
Potassium permanganate solution	0.5% aqueous KMnO4	Exp. 1
Sulfuric acid, acetic acid, hydrochloric acid	10% H2SO4; dilute CH3COOH; dilute HCl	Exp. 1, 8
Uffelmann's reagent	1% phenol + 1–2 drops of 1% FeCl3 (violet), freshly prepared	Exp. 2
Iodine–KI solution + NaOH	Lugol's solution; 10% NaOH	Exp. 2
Potassium hydroxide solution	10% aqueous KOH	Exp. 3
Copper(II) sulfate + sodium hydroxide	5% CuSO4; 10% NaOH	Exp. 3
Lime water	saturated Ca(OH)2 solution	Exp. 4
Iron(III) chloride solution	1% aqueous FeCl3	Exp. 5, 6
Bromine water	saturated aqueous Br2	Exp. 5, 7
Sodium hydroxide / sodium hydrogen carbonate	10% NaOH; 5% NaHCO3	Exp. 8

Equipment: test tubes and rack, pipettes, water bath, glass rods, indicator paper, fume hood, safety goggles and gloves.

6. Laboratory Experiments

Experiment 1 — Reactions of Oxalic Acid

Principle: oxalate ions form a white precipitate of calcium oxalate, insoluble in acetic acid but soluble in mineral acids; oxalic acid is oxidised by permanganate in acidic medium to CO2.

Procedure:

1. To 1 mL of oxalic acid solution add 1 mL of CaCl2 solution. Divide the precipitate into two tubes; add acetic acid to one and dilute HCl to the other.
2. To 1 mL of oxalic acid solution add 1 mL of 10% H2SO4 and 2–3 drops of KMnO4 solution. Warm the tube in a water bath (60–70 °C) and observe.

$$H2C2O4 + CaCl2 \rightarrow CaC2O4\downarrow + 2HCl$$

$$5H2C2O4 + 2KMnO4 + 3H2SO4 \rightarrow 10CO2\uparrow + 2MnSO4 + K2SO4 + 8H2O$$

Result: a white fine-crystalline precipitate of calcium oxalate that does not dissolve in acetic acid but dissolves in HCl; the violet colour of permanganate disappears (slowly at first, then quickly — autocatalysis by Mn2+) with evolution of gas bubbles (CO2).

Experiment 2 — Reactions of Lactic Acid

Principle: (a) Uffelmann's test: the violet iron(III) phenolate complex is destroyed by lactic acid, which binds Fe3+ as yellow-green iron(III) lactate; (b) the CH3–CH(OH)– fragment is oxidised and iodinated in alkaline medium to give iodoform.

Procedure:

1. Uffelmann's test: add 1–2 drops of 1% FeCl3 to 2 mL of 1% phenol solution (violet colour), then add lactic acid solution dropwise.
2. Iodoform test: to 1 mL of lactic acid solution add 1–2 mL of 10% NaOH, then iodine–KI solution dropwise until a faint yellow colour persists; warm gently to 40–50 °C.

$$CH3–CH(OH)–COOH + I2 / NaOH \rightarrow CHI3\downarrow + NaOOC–COONa \quad (schematic)$$

Result: (a) the violet colour changes to yellow-green; (b) a yellow precipitate of iodoform with its characteristic odour.

Note: Uffelmann's test was used historically to detect lactic acid in gastric juice.

Experiment 3 — Reactions of Tartaric Acid

Principle: (a) potassium hydrogen tartrate is poorly soluble in water and precipitates as white crystals; (b) tartrate, a vicinal diol, forms a soluble deep-blue chelate with Cu(OH)2 in alkaline medium.

Procedure:

1. To 1 mL of tartaric acid solution add 10% KOH dropwise, shaking vigorously and rubbing the tube wall with a glass rod. After crystals appear, continue adding KOH in excess.
2. In a second tube, prepare Cu(OH)2 by mixing 1 mL of CuSO4 solution with 2 mL of 10% NaOH (blue precipitate). Add tartaric acid solution dropwise with shaking.

$$HOOC–CH(OH)–CH(OH)–COOH + KOH \rightarrow KOOC–CH(OH)–CH(OH)–COOH\downarrow + H2O$$

$$KOOC–CH(OH)–CH(OH)–COOH + KOH \rightarrow KOOC–CH(OH)–CH(OH)–COOK + H2O$$

$$tartrate + Cu(OH)2\ (NaOH) \rightarrow soluble\ deep\text{-}blue\ copper(II)\ tartrate\ chelate$$

Result: (a) a white crystalline precipitate of potassium hydrogen tartrate, which dissolves in excess KOH (formation of the neutral salt); (b) the blue precipitate of Cu(OH)2 dissolves, giving a clear deep-blue solution — the principle of Fehling's solution.

Experiment 4 — Distinguishing Citric Acid from Tartaric Acid with Calcium Ions

Principle: calcium citrate is less soluble in hot water than in cold: it does not precipitate from a cold solution but separates on boiling and redissolves on cooling. Calcium tartrate precipitates in the cold.

Procedure:

1. To 1 mL of citric acid solution add lime water in excess (until alkaline to indicator paper, about 3–5 mL). Note whether a precipitate forms in the cold.
2. Heat the mixture to boiling and observe; then cool the tube under running water.
3. Repeat the test with tartaric acid solution for comparison.

$$2\ C6H8O7 + 3Ca(OH)2 \rightarrow Ca3(C6H5O7)2\downarrow + 6H2O \quad (on\ boiling)$$

Result: citric acid — the solution stays clear in the cold, a white precipitate of calcium citrate forms on boiling and dissolves again on cooling; tartaric acid — a white precipitate of calcium tartrate forms already in the cold.

Experiment 5 — Keto–Enol Tautomerism of Ethyl Acetoacetate

Principle: the enol form gives a red-violet complex with Fe3+; bromine adds rapidly to the C=C bond of the enol and the colour disappears; the equilibrium then restores the enol form and the colour returns.

Procedure:

1. Shake 2–3 drops of ethyl acetoacetate with 2 mL of water and add 1–2 drops of FeCl3 solution.
2. Add bromine water dropwise with shaking until the colour just disappears. Observe the tube for 1–2 minutes.
3. Repeat the addition of bromine water several times.

$$CH3–CO–CH2–COOC2H5 \rightleftharpoons CH3–C(OH)=CH–COOC2H5$$

$$CH3–C(OH)=CH–COOC2H5 + Br2 \rightarrow CH3–CO–CHBr–COOC2H5 + HBr$$

Result: a red-violet colour with FeCl3; it disappears after addition of bromine water and reappears within a short time; the cycle can be repeated several times — evidence of a dynamic keto–enol equilibrium.

Experiment 6 — Reaction of Phenols with Iron(III) Chloride

Principle: phenols form coloured complexes (iron(III) phenolates) with Fe3+; the colour depends on the number and position of the OH groups.

Procedure:

1. Place 1 mL of the solutions of phenol, resorcinol, hydroquinone and salicylic acid in four test tubes, and 1 mL of ethanol in a fifth tube as a control.
2. Add 1–2 drops of 1% FeCl3 solution to each tube and compare the colours.

$$6\ C6H5OH + FeCl3 \rightarrow H3[Fe(OC6H5)6] + 3HCl \quad (simplified)$$

Result: phenol — violet; resorcinol — dark violet; salicylic acid — violet (red-violet); hydroquinone — transient greenish colour, then yellow (oxidation to p-benzoquinone), sometimes dark-green crystals of quinhydrone; ethanol — no change.

Experiment 7 — Bromination of Phenol

Principle: the OH group strongly activates the ring, so bromine water brominates phenol at all three ortho/para positions without a catalyst.

Procedure: to 1 mL of phenol solution add bromine water dropwise with shaking.

$$C6H5OH + 3Br2 \rightarrow 2,4,6\text{-}Br3C6H2OH\downarrow + 3HBr$$

Result: bromine water is decolourised immediately and a white precipitate of 2,4,6-tribromophenol forms.

Experiment 8 — Acidic Properties of Phenol Compared with a Carboxylic Acid

Principle: phenol (pKa ≈ 10) reacts with NaOH but is too weak an acid to decompose NaHCO3 (carbonic acid pKa1 = 6.4); carboxylic acids react with both.

Procedure:

1. Place 1 mL of phenol emulsion (a turbid mixture of phenol and water) in a test tube and add 10% NaOH dropwise until the solution becomes clear.
2. Add dilute HCl to the clear solution until it is acidic and observe.
3. Add 1 mL of 5% NaHCO3 to 1 mL of phenol emulsion, and in another tube to a few crystals of tartaric or salicylic acid. Compare.

$$C6H5OH + NaOH \rightarrow C6H5ONa + H2O$$

$$C6H5ONa + HCl \rightarrow C6H5OH + NaCl$$

$$C6H5OH + NaHCO3 \rightarrow no\ reaction$$

Result: phenol dissolves in NaOH (sodium phenoxide) and separates again as an emulsion on acidification; no gas evolves with NaHCO3, whereas the carboxylic acid releases CO2.

7. Safety Precautions

- Phenol and its solutions are toxic and cause severe chemical burns; phenol is rapidly absorbed through the skin. Always wear gloves. If phenol contacts the skin, wash it off immediately with plenty of water (or polyethylene glycol 300/400, if available) and inform the teacher.
- Bromine water releases toxic, irritating bromine vapour — work in the fume hood.
- Oxalic acid and its salts are toxic if swallowed (they bind calcium).
- Potassium permanganate is a strong oxidiser and stains skin; NaOH and KOH solutions are caustic.
- Wear a lab coat, safety goggles, and gloves at all times; no food or drink in the laboratory.
- Know the location of the eyewash station, safety shower, and fire extinguisher before starting work.
- Dispose of phenol- and bromine-containing waste in the designated containers.

8. Individual Assignment and Report Requirements

Individual assignment: a set of unlabelled test tubes contains solutions of oxalic acid, lactic acid, tartaric acid, citric acid and phenol. Propose a logical scheme of reactions from this work that identifies each substance, and write the equations of all the reactions you use.

Protocol (laboratory report) requirements. For each experiment performed, the notebook must contain:

1. Name of the experiment and the reagents used.
2. Balanced equation(s) of the reactions.
3. Observations (colour, precipitate, gas, odour, mass, melting or boiling point) described precisely, not just “positive/negative”.
4. Calculations where required (theoretical and percent yield, specific rotation, etc.).
5. A short conclusion for each experiment linking the result to the structure of the compound.

Complete the table below with your observations and conclusions.

Substance	Reagent / test	Observation	Conclusion (functional group or feature)
Oxalic acid	CaCl2; KMnO4 / H2SO4		
Lactic acid	Uffelmann's reagent; I2 / NaOH		
Tartaric acid	KOH; Cu(OH)2		
Citric acid	Ca(OH)2, boiling		
Ethyl acetoacetate	FeCl3; bromine water		
Phenol	FeCl3; bromine water; NaOH / NaHCO3		
Resorcinol, hydroquinone, salicylic acid	FeCl3		

9. Recommended Literature

1. Clayden, J., Greeves, N., Warren, S. Organic Chemistry. Oxford University Press.
2. Morrison, R. T., Boyd, R. N. Organic Chemistry.
3. Furniss, B. S. et al. (Vogel). Vogel's Textbook of Practical Organic Chemistry.
4. Ahluwalia, V. K., Aggarwal, R. Comprehensive Practical Organic Chemistry: Preparations and Qualitative Analysis.
5. The official study guide and lecture notes for Chemistry of the Functional Derivatives of Organic Molecules (current program, Year 3).

Students should also consult the current national/institutional pharmacopoeia for official identification reactions and assay methods related to this topic.

Laboratory work No. 11

Amino Acids. Peptides and Proteins. Qualitative Reactions

Topic: Structure and properties of α-amino acids. The peptide bond and levels of protein structure. Qualitative and precipitation reactions of amino acids and proteins

1. Relevance of the Topic

α-Amino acids are the building blocks of proteins and the precursors of many biologically active substances — neurotransmitters, hormones and coenzymes. Amino acids themselves are used as drugs (glycine, glutamic acid, methionine, acetylcysteine), and peptides and proteins form a rapidly growing class of therapeutics (insulin, oxytocin, monoclonal antibodies).

Qualitative reactions of amino acids and proteins are used to identify and assay these substances in pharmaceutical analysis, and in clinical laboratories to detect protein in biological fluids. Understanding precipitation and denaturation is important for the formulation and storage of protein drugs and explains, for example, why milk and egg white are given as first aid in poisoning with heavy-metal salts.

2. Learning Objectives

By the end of the session, the student must know:

- The structure, classification and stereochemistry (L-series) of α-amino acids.
- The amphoteric properties of amino acids: the zwitterion and the isoelectric point.
- Reactions of amino acids as amines and as acids; formation of chelate complexes.
- The structure of the peptide bond and the four levels of protein structure.
- The chemistry of the biuret, ninhydrin, xanthoproteic and Fohl reactions and of protein precipitation reactions.

By the end of the session, the student must be able to:

- Detect free α-amino acids and peptide bonds in a sample.
- Detect aromatic and sulfur-containing amino acid residues in proteins.
- Distinguish reversible precipitation (salting out) from irreversible denaturation.
- Explain each observation with an equation or a structural argument.

3. Theoretical Background

3.1 α-Amino acids

α-Amino acids have the general formula R–CH(NH2)–COOH; twenty of them (the proteinogenic amino acids) are encoded by the genetic code. All except glycine are chiral, and natural amino acids belong to the L-series (in the CIP system mostly S; cysteine is R).

Amino acids are amphoteric. In the solid state and in neutral solution they exist as zwitterions (inner salts) H3N+–CH(R)–COO−, which explains their high melting points and solubility in water. The pH at which the net charge of the molecule is zero is the isoelectric point, pI (glycine pI = 5.97). As bifunctional compounds, amino acids react both as amines (acylation, reaction with nitrous acid, Schiff base

formation) and as acids (salt formation, esterification), and form stable chelate complexes with metal ions, such as the deep-blue copper(II) glycinate.

The reaction with ninhydrin is a general test for α-amino acids. The amino acid is oxidatively deaminated and decarboxylated, and the released ammonia condenses with a second ninhydrin molecule to form a blue-violet dye (Ruhemann's purple). Proline, a secondary amine, gives a yellow colour.

R–CH(NH2)–COOH + 2 ninhydrin → Ruhemann's purple + R–CHO + CO2 + 3H2O (schematic)

3.2 Peptides and proteins

In peptides and proteins, amino acid residues are linked by peptide (amide) bonds –CO–NH–. The peptide bond is planar and has partial double-bond character because the nitrogen lone pair is conjugated with the C=O group; rotation about it is restricted, and it usually adopts the trans configuration.

Protein structure is described at four levels: primary (the amino acid sequence), secondary (α-helix and β-sheet, stabilised by hydrogen bonds between peptide groups), tertiary (the overall folding of one chain, stabilised by disulfide bridges, ionic, hydrogen and hydrophobic interactions) and quaternary (association of several subunits, as in haemoglobin).

Qualitative reactions of proteins include (1) the general biuret reaction for peptide bonds and (2) colour reactions for particular residues: the xanthoproteic reaction for aromatic residues (tyrosine, tryptophan), the Fohl reaction for sulfur-containing residues (cysteine, cystine), Millon's reaction for tyrosine and the Sakaguchi reaction for arginine (the last two are not performed in this work). Proteins are also characterised by precipitation reactions: reversible (salting out with neutral salts) and irreversible (denaturation by heat, heavy-metal salts, concentrated acids and organic solvents).

4. Self-Control Questions (Pre-lab Preparation)

1. Write the zwitterionic form of glycine and explain why amino acids have high melting points and are soluble in water.
2. What is the isoelectric point? In which direction will glycine move in an electric field at pH 2 and at pH 10?
3. Write the equation of the formation of copper(II) glycinate. What type of compound is it?
4. Explain the principle of the ninhydrin reaction. Why does proline give a different colour?
5. Describe the structure of the peptide bond. Why is it planar?
6. Which structural element of proteins is detected by the biuret reaction? Why do free amino acids give a negative result?
7. Which amino acid residues give a positive xanthoproteic reaction? Explain the change of colour on addition of alkali.
8. Why does gelatin give a weak xanthoproteic reaction compared with egg albumin?
9. Which amino acids are detected by the Fohl reaction? Why is methionine negative?
10. What is the difference between salting out and denaturation? Give examples of reagents causing each.

5. Reagents, Materials, and Equipment

Test substances: egg white solution (egg white diluted 1:10 with water and filtered through gauze), 1% gelatin solution, 1% glycine solution, cysteine solution (if available).

Reagent	Composition / notes	Used in
Copper(II) sulfate solutions	1% and 5% aqueous CuSO4	Exp. 1, 3, 6
Sodium hydroxide solutions	10% and 30% aqueous NaOH	Exp. 1, 3, 4, 5
Ninhydrin solution	0.1–0.2% in water or ethanol	Exp. 2
Concentrated nitric acid	HNO3, 65–68%	Exp. 4
Lead(II) acetate solution	5% aqueous Pb(CH3COO)2	Exp. 5
Ammonium sulfate	saturated solution and solid (NH4)2SO4	Exp. 6
Acetic acid	1% aqueous CH3COOH	Exp. 6
Ethanol	96%	Exp. 6

Equipment: test tubes and rack, pipettes, boiling water bath, hot plate, filter funnels and filter paper, glass rods, fume hood, safety goggles and gloves.

6. Laboratory Experiments

Experiment 1 — Formation of Copper(II) Glycinate (Chelate Complex)

Principle: α-amino acids act as bidentate ligands (nitrogen of the amino group and oxygen of the carboxylate) and form stable, intensely coloured chelate complexes with Cu2+.

Procedure:

1. Prepare a small amount of Cu(OH)2: mix 2–3 drops of 5% CuSO4 with 3–4 drops of 10% NaOH (blue precipitate).
2. Add 1–2 mL of glycine solution and shake.
3. For comparison, add 1–2 mL of water to a second portion of Cu(OH)2.

$$2\ H2N–CH2–COOH + Cu(OH)2 \rightarrow (H2N–CH2–COO)2Cu + 2H2O$$

Result: the precipitate dissolves and a dark-blue solution of copper(II) glycinate forms; with water the precipitate remains unchanged.

Experiment 2 — Ninhydrin Reaction

Principle: free α-amino groups react with ninhydrin on heating to form the blue-violet Ruhemann's purple.

Procedure:

1. Place 1 mL of glycine solution in one test tube and 1 mL of egg white solution in another.
2. Add 4–5 drops of ninhydrin solution to each tube and heat in a boiling water bath for 1–2 minutes.

Result: a blue-violet colour develops with glycine and with the protein solution (free α-amino groups of the amino acid and of the protein).

Experiment 3 — Biuret Reaction (Test for Peptide Bonds)

Principle: in strongly alkaline medium, Cu2+ forms violet complexes with the nitrogen atoms of peptide groups. At least two peptide bonds are required, so free amino acids do not give the violet colour.

Procedure:

1. Place 1 mL of egg white solution, 1 mL of gelatin solution and 1 mL of glycine solution in three test tubes.
2. Add 1 mL of 10% NaOH to each tube, then 1–2 drops of 1% $CuSO_4$, and shake. Avoid excess $CuSO_4$ — its blue colour masks the reaction.

Result: egg white and gelatin — violet (red-violet) colour; glycine — blue colour only (negative).

Experiment 4 — Xanthoproteic Reaction (Aromatic Amino Acid Residues)

Principle: concentrated nitric acid nitrates the activated aromatic rings of tyrosine and tryptophan residues, forming yellow nitro derivatives; in alkaline medium they are converted into orange salts.

Procedure:

1. In the fume hood, add 5–6 drops of concentrated HNO_3 to 1 mL of egg white solution (a white precipitate of denatured protein forms) and to 1 mL of gelatin solution in another tube.
2. Heat carefully in a water bath for 1–2 minutes and observe the colour.
3. Cool the tubes and carefully add 30% NaOH dropwise until alkaline.

$$\text{tyrosine residue} + HNO_3 \rightarrow \text{3-nitrotyrosine residue (yellow)} + H_2O;\ \text{in NaOH} \rightarrow \text{orange phenolate}$$

Result: egg white — the precipitate and solution turn yellow on heating and orange after addition of alkali; gelatin — only a weak pale-yellow colour, because gelatin contains little tyrosine and no tryptophan.

Note: the yellow staining of skin by nitric acid is a xanthoproteic reaction of skin proteins.

Experiment 5 — Fohl Reaction (Sulfur-Containing Amino Acid Residues)

Principle: on boiling with concentrated alkali, the sulfur of cysteine and cystine residues is split off as sulfide ion, which gives a black-brown precipitate of lead(II) sulfide with lead(II) ions. Methionine does not react because its sulfur is bound more firmly (thioether).

Procedure:

1. In the fume hood, to 1 mL of egg white solution add an equal volume of 30% NaOH and 3–4 drops of lead(II) acetate solution.
2. Heat the mixture carefully to boiling and boil for 2–3 minutes; let it stand.
3. Repeat the test with gelatin solution (and cysteine solution, if available).

$$R\text{–}CH_2\text{–}SH + 2NaOH \rightarrow R\text{–}CH_2\text{–}OH + Na_2S + H_2O \quad \text{(schematic)}$$

$$Pb(CH_3COO)_2 + 4NaOH \rightarrow Na_2[Pb(OH)_4] + 2CH_3COONa$$

$$Na_2S + Na_2[Pb(OH)_4] \rightarrow PbS\downarrow + 4NaOH$$

Result: egg white — brown to black precipitate (PbS); gelatin — negative or very weak (gelatin is poor in cysteine).

Experiment 6 — Precipitation of Proteins (Salting Out and Denaturation)

Principle: neutral salts at high concentration remove the hydration shell and charge of protein molecules and precipitate them reversibly (salting out); heat, heavy-metal ions and organic solvents destroy the native structure (denaturation) and precipitate proteins irreversibly.

Procedure:

1. Salting out: to 2 mL of egg white solution add an equal volume of saturated (NH4)2SO4 solution (globulins precipitate). Filter and saturate the filtrate with solid (NH4)2SO4 (albumins precipitate). Dilute part of the precipitate with water.
2. Heat coagulation: acidify 2 mL of egg white solution with 1–2 drops of 1% acetic acid and heat to boiling.
3. Heavy-metal salts: add 5% CuSO4 dropwise to 1 mL of egg white solution, then add excess CuSO4.
4. Organic solvent: add 2 mL of ethanol to 1 mL of egg white solution.

Result: (1) a white precipitate that redissolves on dilution with water — reversible; (2) white flakes of coagulated protein; (3) a precipitate that may partly redissolve in excess CuSO4 (adsorption of ions); (4) turbidity and a precipitate. Precipitation in (2)–(4) is irreversible (denaturation).

Practical significance: milk and egg white are given as first aid in poisoning with heavy-metal salts, because their proteins bind the metal ions.

7. Safety Precautions

- Concentrated HNO3 and 30% NaOH are highly corrosive. Heat test tubes containing them in the fume hood, pointing the opening away from yourself and others.
- Lead(II) acetate is toxic (heavy metal, reproductive toxin). Use small amounts, wear gloves, and collect all lead-containing waste in the designated container.
- Ninhydrin is harmful and stains skin purple — avoid contact.
- Copper salts are harmful if swallowed; ethanol is flammable — keep it away from flames and hot plates.
- Wear a lab coat, safety goggles, and gloves at all times; no food or drink in the laboratory.
- Know the location of the eyewash station, safety shower, and fire extinguisher before starting work.

8. Individual Assignment and Report Requirements

Individual assignment: an unknown solution contains either a protein or a free amino acid. Propose a sequence of tests from this work that establishes which it is and, for a protein, whether it contains tyrosine/tryptophan and cysteine residues. Write the structure of the tripeptide Gly–Tyr–Cys and state which tests it would give.

Protocol (laboratory report) requirements. For each experiment performed, the notebook must contain:

1. Name of the experiment and the reagents used.
2. Balanced equation(s) of the reactions.
3. Observations (colour, precipitate, gas, odour, mass, melting or boiling point) described precisely, not just “positive/negative”.
4. Calculations where required (theoretical and percent yield, specific rotation, etc.).
5. A short conclusion for each experiment linking the result to the structure of the compound.

Complete the table with the results (“+” or “–” and the colour) for each test substance.

Test	Egg white	Gelatin	Glycine	Detected structure
Cu(OH)2 (chelate)				α-amino acid
Ninhydrin				free α-NH2 group
Biuret				peptide bonds (≥ 2)
Xanthoproteic				aromatic residues (Tyr, Trp)
Fohl				cysteine, cystine
Heat / CuSO4 / ethanol				protein (denaturation)

9. Recommended Literature

1. Nelson, D. L., Cox, M. M. Lehninger Principles of Biochemistry.
2. Plummer, D. T. An Introduction to Practical Biochemistry.
3. Clayden, J., Greeves, N., Warren, S. Organic Chemistry. Oxford University Press.
4. Morrison, R. T., Boyd, R. N. Organic Chemistry.
5. The official study guide and lecture notes for Chemistry of the Functional Derivatives of Organic Molecules (current program, Year 3).

Students should also consult the current national/institutional pharmacopoeia for official identification reactions and assay methods related to this topic.

Laboratory work No. 12

Methods of Derivatization of Enantiomers. Common Derivatizing Reagents

Topic: Chirality, enantiomers and diastereomers. Indirect separation and analysis of enantiomers by chiral derivatization. Resolution of a racemate via diastereomeric salts

1. Relevance of the Topic

Most modern drug substances are chiral, and their enantiomers often differ greatly in pharmacological activity, metabolism and toxicity: (S)-ibuprofen is the active anti-inflammatory enantiomer and (S)-propranolol is the active β-blocker. The history of thalidomide showed how important stereochemistry is for drug safety. Pharmacopoeias and regulatory agencies therefore require the enantiomeric purity of chiral drugs to be determined.

Enantiomers have identical physical properties in an achiral environment and cannot be separated or distinguished by ordinary chromatography or NMR. The classical solution is derivatization: reaction with an enantiomerically pure reagent converts a pair of enantiomers into a pair of diastereomers, which differ in physical properties. They can then be separated by crystallisation or achiral chromatography, or distinguished by NMR. This work introduces the principles of this approach and the most widely used chiral derivatizing reagents.

2. Learning Objectives

By the end of the session, the student must know:

- The concepts of chirality, stereogenic centre, enantiomers, diastereomers, racemate, enantiomeric excess and optical purity.
- Why enantiomers cannot be separated by achiral methods, while diastereomers can.
- The principle of indirect enantiomer analysis with chiral derivatizing agents (CDAs) and the requirements for a good CDA.
- The most common derivatizing reagents, the functional groups they react with, and the analytical methods used for the derivatives.
- The difference between indirect (derivatization) and direct (chiral stationary phases, chiral solvating agents) methods.

By the end of the session, the student must be able to:

- Measure optical rotation and calculate specific rotation and concentration.
- Carry out the resolution of a racemic amine via diastereomeric salts and liberate the resolved enantiomer.
- Calculate enantiomeric excess (optical purity) from polarimetric data.
- Choose a suitable derivatizing reagent for a given analyte and write the derivatization reaction.

3. Theoretical Background

3.1 Enantiomers, diastereomers and their properties

A chiral molecule is not superimposable on its mirror image; the two mirror-image forms are enantiomers. In an achiral environment, enantiomers have identical melting and boiling points, solubility, NMR and IR spectra and chromatographic behaviour. They differ only in the direction in which they rotate plane-polarised light and in their interaction with other chiral objects (receptors, enzymes, chiral reagents and sorbents). Diastereomers — stereoisomers that are not mirror images — differ in all physical properties.

An equimolar mixture of enantiomers (a racemate) is optically inactive. The composition of a mixture of enantiomers is expressed as enantiomeric excess (ee), and the optical rotation of a substance is characterised by its specific rotation:

$$ee\ (\%) = ([R] - [S]) / ([R] + [S]) \times 100$$

$$[\alpha] = \alpha / (l \cdot c) \quad (\alpha \text{ — observed rotation, °; } l \text{ — tube length, dm; } c \text{ — concentration, g/mL})$$

$$\text{optical purity}\ (\%) = [\alpha]_{observed} / [\alpha]_{pure\ enantiomer} \times 100 \approx ee$$

3.2 The principle of chiral derivatization

In the indirect method, a mixture of enantiomers (R)-A and (S)-A reacts with a single, enantiomerically pure reagent (R)-B. Two diastereomers are formed, (R)-A–(R)-B and (S)-A–(R)-B. They can be separated by crystallisation or on an ordinary (achiral) GC or HPLC column, or distinguished by NMR because their signals have different chemical shifts. The ratio of the diastereomers equals the original ratio of the enantiomers.

$$(R)\text{-}A + (S)\text{-}A + 2\ (R)\text{-}B \rightarrow (R)\text{-}A\text{–}(R)\text{-}B + (S)\text{-}A\text{–}(R)\text{-}B \quad \text{(diastereomers)}$$

Requirements for a chiral derivatizing agent: it must be enantiomerically pure (traces of its enantiomer distort the result); it must react quantitatively and at the same rate with both enantiomers (no kinetic resolution); neither the analyte nor the reagent may racemise during the reaction; and the derivatives must be stable and easy to detect (UV-absorbing or fluorescent groups, a 19F label for NMR).

Derivatization can also be preparative. In the classical resolution of racemates (Pasteur, 1853), a racemic acid or base is converted into diastereomeric salts with an enantiomerically pure base or acid (brucine, quinine, 1-phenylethylamine, tartaric acid, camphor-10-sulfonic acid). The salts differ in solubility and are separated by fractional crystallisation; the pure enantiomer is then liberated with a strong acid or base. This approach is used in the present work.

3.3 Common chiral derivatizing reagents

Reagent	Analyte (functional group)	Diastereomeric derivative	Typical analysis
Mosher's acid chloride, MTPA-Cl ((R)- or (S)-α-methoxy-α-trifluoromethylphenylacetyl chloride)	alcohols, amines	MTPA esters / amides	1H and 19F NMR; absolute configuration; GC, HPLC
Marfey's reagent, FDAA (1-fluoro-2,4-dinitrophenyl-5-L-alanine amide)	amino acids, amines	dinitrophenyl-alanine amide derivatives	reversed-phase HPLC, UV 340 nm

Reagent	Analyte (functional group)	Diastereomeric derivative	Typical analysis
GITC (2,3,4,6-tetra-O-acetyl-β-D-glucopyranosyl isothiocyanate)	amines, amino acids, β-blockers	thioureas	reversed-phase HPLC, UV
o-Phthalaldehyde + chiral thiol (N-acetyl-L-cysteine)	primary amines, amino acids	isoindoles	HPLC, fluorescence
(+)- or (−)-FLEC (1-(9-fluorenyl)ethyl chloroformate)	amines, amino acids	carbamates	HPLC, fluorescence
(−)-Menthyl chloroformate	amines, alcohols	carbamates / carbonates	GC, HPLC
(−)-Camphanic acid chloride	alcohols, amines	esters / amides	HPLC, NMR, X-ray
(R)- or (S)-1-Phenylethylamine; (S)-1-(1-naphthyl)ethylamine	carboxylic acids (e.g. profens)	amides	HPLC, GC
(R)- or (S)-1-(1-Naphthyl)ethyl isocyanate	alcohols, amines	carbamates / ureas	HPLC, UV / fluorescence
L-(+)-Tartaric acid, (+)-camphor-10-sulfonic acid, brucine	bases / acids	diastereomeric salts	fractional crystallisation (preparative resolution)

Direct methods do not require covalent derivatization: enantiomers are separated on chiral stationary phases (cyclodextrin-, polysaccharide- or protein-based columns), by capillary electrophoresis with chiral selectors, or distinguished by NMR in the presence of chiral solvating agents or chiral lanthanide shift reagents (for example, Eu(hfc)3).

4. Self-Control Questions (Pre-lab Preparation)

1. Define enantiomers, diastereomers and a racemate. Which physical properties do enantiomers share, and in which do they differ?
2. Why can enantiomers not be separated by ordinary crystallisation or on an achiral chromatographic column?
3. Explain the principle of the indirect method of enantiomer analysis. Why must the derivatizing agent be enantiomerically pure?
4. What is kinetic resolution, and why does it cause errors in the determination of enantiomeric composition?
5. Which reagents are used to derivatize (a) chiral alcohols, (b) amino acids, (c) chiral carboxylic acids? Write one reaction for each.
6. Why does Mosher's reagent contain a CF3 group? Which analytical method makes use of it?
7. Write the equation for the formation of diastereomeric salts of racemic 1-phenylethylamine with L-(+)-tartaric acid.
8. How is the free amine recovered from its tartrate salt?
9. A solution of a chiral compound (c = 0.050 g/mL) in a 1 dm tube shows α = +2.1°. Calculate its specific rotation.
10. A sample of (S)-1-phenylethylamine has [α] = −32°; the pure enantiomer has [α] = −40°. Calculate the optical purity and the percentage of each enantiomer.

5. Reagents, Materials, and Equipment

Test substances: sucrose (analytical grade), racemic 1-phenylethylamine (α-methylbenzylamine), L-(+)-tartaric acid.

Reagent	Composition / notes	Used in
Sucrose solutions	standard solution 0.10 g/mL and a solution of unknown concentration	Exp. 1
L-(+)-Tartaric acid	(2R,3R)-tartaric acid, solid	Exp. 2
Racemic 1-phenylethylamine	(±)-C6H5CH(CH3)NH2, liquid (d 0.95 g/mL)	Exp. 2
Methanol	solvent for crystallisation	Exp. 2, 3
Sodium hydroxide solution	50% aqueous NaOH	Exp. 2
Diethyl ether or tert-butyl methyl ether	extraction solvent	Exp. 2
Anhydrous sodium sulfate	drying agent	Exp. 2

Equipment: polarimeter with 1 dm and 2 dm tubes, analytical balance, volumetric flasks (10–50 mL), conical flasks (125 mL), water bath, Büchner funnel with filter flask, separating funnel, beakers, fume hood, safety goggles and gloves.

6. Laboratory Experiments

Experiment 1 — Measurement of Optical Rotation. Specific Rotation of Sucrose

Principle: the angle of rotation of plane-polarised light is proportional to the concentration of the optically active substance and to the length of the tube. From the measured angle, the specific rotation — a characteristic constant — or an unknown concentration can be calculated.

Procedure:

1. Check the zero point of the polarimeter with the tube filled with distilled water.
2. Fill a 2 dm tube with the standard sucrose solution (0.10 g/mL), avoiding air bubbles, and measure the angle of rotation three times; take the mean value.
3. Calculate the specific rotation of sucrose and compare it with the literature value.
4. Measure the rotation of the sucrose solution of unknown concentration in the same way and calculate its concentration.

$$[\alpha] = \alpha / (l \cdot c); \quad c = \alpha / ([\alpha] \cdot l)$$

Result: for c = 0.10 g/mL and l = 2 dm, α ≈ +13.3°, which gives [α]D20 ≈ +66.5° (the literature value for sucrose).

Experiment 2 — Resolution of Racemic 1-Phenylethylamine via Diastereomeric Salts

Principle: racemic 1-phenylethylamine reacts with enantiomerically pure (R,R)-tartaric acid to form two diastereomeric salts. In methanol, the salt of the (S)-amine is less soluble and crystallises, while the salt of the (R)-amine stays in solution. Treatment of the crystals with alkali liberates the (S)-(−)-amine.

Procedure:

1. In a 125 mL conical flask, dissolve 3.9 g (26 mmol) of L-(+)-tartaric acid in 55 mL of methanol by heating on a water bath almost to boiling.

2. In the fume hood, slowly and carefully add 3.1 g (3.3 mL, 26 mmol) of racemic 1-phenylethylamine to the hot solution in small portions (the reaction is exothermic).
3. Stopper the flask and leave it to crystallise undisturbed at room temperature until the next session (at least 24 h). The (S)-amine salt crystallises as prisms; if needles form instead, redissolve them by heating and allow the solution to cool slowly.
4. Collect the crystals by vacuum filtration, wash them with a small amount of cold methanol, dry in air and weigh.
5. Dissolve the salt in about 15 mL of water and add 3 mL of 50% NaOH (the solution must be strongly alkaline): the free amine separates as an oily layer.
6. Extract the amine with two 10 mL portions of ether, combine the extracts and dry them over anhydrous Na2SO4.
7. Decant the solution into a pre-weighed flask and evaporate the ether on a warm water bath in the fume hood (no open flames!). Weigh the residual amine.

   (R,S)-C6H5CH(CH3)NH2 + (R,R)-tartaric acid → (S)-amine · (R,R)-hydrogen tartrate↓ (less soluble) + (R)-amine salt (in solution)

   (S)-C6H5CH(CH3)NH3+ HT− + 2NaOH → (S)-C6H5CH(CH3)NH2 + Na2T + 2H2O (T = tartrate)

Result: colourless prismatic crystals of the diastereomeric salt; after liberation, a colourless liquid amine with an ammonia-like odour, enriched in the (S)-(−)-enantiomer. The maximum amount of one enantiomer is 1.55 g (half of the racemate).

Experiment 3 — Determination of the Optical Purity of the Resolved Amine

Principle: the optical purity of the resolved amine is found by comparing its specific rotation with that of the pure enantiomer ((S)-(−)-1-phenylethylamine, [α]D ≈ −40°, neat liquid).

Procedure:

1. Measure the rotation of the neat amine in a short polarimeter tube (for the neat liquid, c = density = 0.95 g/mL). If the amount is insufficient, dissolve a weighed sample in methanol in a volumetric flask and compare with the reference value for the same solvent provided by the teacher.
2. Calculate the specific rotation, the optical purity and the percentage of each enantiomer.

   optical purity (%) = [α]observed / (−40°) × 100

   % (S) = (100 + ee) / 2; % (R) = (100 − ee) / 2

Result: a single crystallisation usually gives an amine of moderate to high optical purity; recrystallisation of the salt increases it further.

7. Safety Precautions

- 1-Phenylethylamine is corrosive and harmful; it causes skin burns and eye damage. Handle it only in the fume hood with gloves and goggles.
- Methanol is toxic by inhalation, ingestion and skin absorption, and flammable. Diethyl ether is extremely flammable and forms explosive peroxides on storage. No open flames; evaporate solvents only on a water bath in the fume hood.
- 50% NaOH is highly caustic.
- Handle polarimeter tubes carefully and clean them immediately after use.

- Wear a lab coat, safety goggles, and gloves at all times; no food or drink in the laboratory.
- Know the location of the eyewash station, safety shower, and fire extinguisher before starting work.
- Collect organic solvent waste in the designated container.

8. Individual Assignment and Report Requirements

Individual assignment: for each of the racemic analytes (±)-ibuprofen, (±)-propranolol, DL-alanine and (±)-butan-2-ol, choose a suitable chiral derivatizing reagent from the table in section 3.3, write the derivatization reaction, and name the analytical method you would use to separate or distinguish the diastereomers.

Protocol (laboratory report) requirements. For each experiment performed, the notebook must contain:

1. Name of the experiment and the reagents used.
2. Balanced equation(s) of the reactions.
3. Observations (colour, precipitate, gas, odour, mass, melting or boiling point) described precisely, not just "positive/negative".
4. Calculations where required (theoretical and percent yield, specific rotation, etc.).
5. A short conclusion for each experiment linking the result to the structure of the compound.

Record the polarimetric data and calculated values in the table below.

Sample	c (g/mL)	l (dm)	α observed (°)	[α] calculated (°)	Optical purity / ee (%)
Sucrose, standard	0.10	2			—
Sucrose, unknown	to be calculated	2		+66.5 (lit.)	—
Resolved 1-phenylethylamine					

9. Recommended Literature

1. Eliel, E. L., Wilen, S. H., Mander, L. N. Stereochemistry of Organic Compounds. Wiley, 1994.
2. Blau, K., Halket, J. M. (eds.). Handbook of Derivatives for Chromatography. 2nd ed. Wiley, 1993.
3. Dale, J. A., Mosher, H. S. J. Am. Chem. Soc. 1973, 95, 512–519 (MTPA esters for configurational analysis).
4. Marfey, P. Carlsberg Res. Commun. 1984, 49, 591–596 (determination of D-amino acids with FDAA).
5. Clayden, J., Greeves, N., Warren, S. Organic Chemistry. Oxford University Press.
6. The official study guide and lecture notes for Chemistry of the Functional Derivatives of Organic Molecules (current program, Year 3).

Students should also consult the current national/institutional pharmacopoeia for official identification reactions and assay methods related to this topic.

Laboratory work No. 13

Methods for Studying the Spatial Structure of Organic Molecules

Topic: Configuration and conformation. Physical, spectral and chemical methods for determining spatial structure. cis–trans isomerism of maleic and fumaric acids

1. Relevance of the Topic

The biological activity of a drug depends not only on its composition and constitution but also on its three-dimensional structure: the spatial arrangement of functional groups determines how the molecule fits the binding site of a receptor or enzyme. Geometric isomers can differ in activity as strongly as enantiomers — for example, (Z)-tamoxifen is the active antiestrogen, while its (E)-isomer has weak estrogenic activity.

Determining spatial structure is therefore a routine task in drug research and quality control. It is solved by a combination of methods: X-ray crystallography, NMR and optical spectroscopy, polarimetry, dipole moment measurements and chemical correlations. This work gives an overview of these methods and shows, with maleic and fumaric acids as an example, how geometric isomers differ in physical and chemical properties.

2. Learning Objectives

By the end of the session, the student must know:

- The concepts of configuration and conformation; types of stereoisomerism (enantiomers, diastereomers including cis–trans isomers, conformers); R/S and E/Z nomenclature.
- The conformations of ethane, butane and cyclohexane; axial and equatorial positions; relative energies of conformers.
- The physical basis of the main methods for studying spatial structure and the information each method provides.
- How cis–trans isomers differ in physical properties (melting point, solubility, dipole moment) and chemical behaviour.

By the end of the session, the student must be able to:

- Build molecular models of conformers and stereoisomers and represent them with Newman, sawhorse and Fischer projections.
- Carry out the cis → trans isomerisation of maleic acid and identify the product.
- Use NMR coupling constants and IR data to assign the configuration of a double bond or the conformation of a ring substituent.

3. Theoretical Background

3.1 Configuration and conformation

The configuration of a molecule is the spatial arrangement of its atoms that distinguishes stereoisomers and cannot be changed without breaking bonds: the arrangement of substituents about a stereogenic centre (R/S), or about a double bond or ring (cis/trans, E/Z). Conformations are different spatial forms of the

same molecule that interconvert by rotation about single bonds; at room temperature they usually interconvert rapidly and cannot be isolated.

In ethane, the staggered conformation is about 12 kJ/mol more stable than the eclipsed one (torsional strain). In butane, the anti conformation is about 3.8 kJ/mol more stable than the gauche conformation (steric interaction of the methyl groups). Cyclohexane adopts a strain-free chair conformation in which each carbon carries one axial and one equatorial bond; large substituents prefer the equatorial position (about 95% of methylcyclohexane is equatorial at 25 °C).

cis–trans (geometric) isomers are diastereomers and differ in all physical and chemical properties. Maleic acid ((Z)-butenedioic acid) and fumaric acid ((E)-butenedioic acid) are a classic example. Maleic acid is much more soluble in water, melts at a lower temperature, is a stronger acid in the first step (pKa1 1.9 vs 3.0, because an intramolecular hydrogen bond stabilises its monoanion), and forms a cyclic anhydride on heating because its two carboxyl groups are close together. Fumaric acid does not form an anhydride and is thermodynamically more stable.

HOOC–CH=CH–COOH (Z, maleic acid) → HOOC–CH=CH–COOH (E, fumaric acid) (HCl, t)

In the presence of acids (or light with a trace of bromine), the double bond of maleic acid is temporarily converted into a single bond by protonation or reversible addition of a nucleophile. Free rotation then becomes possible, and the molecule returns to the more stable trans form.

3.2 Methods for studying spatial structure

Method	Physical basis	What it establishes	Example
X-ray crystallography	diffraction of X-rays by a crystal lattice	complete 3D structure, bond lengths and angles; absolute configuration (anomalous dispersion)	Bijvoet (1951): absolute configuration of (+)-tartaric acid (as its sodium rubidium salt)
NMR spectroscopy	coupling constants (Karplus relationship), nuclear Overhauser effect, chiral shift reagents	relative configuration, conformation, cis/trans geometry, ee	3J(trans CH=CH) 12–18 Hz vs 3J(cis) 6–12 Hz; 3J(ax,ax) 8–13 Hz in cyclohexanes
Polarimetry, ORD	rotation of plane-polarised light	optical activity, specific rotation, ee	sucrose [α]D +66.5°
Circular dichroism (CD)	unequal absorption of left and right circularly polarised light	absolute configuration (Cotton effect, octant rule); protein secondary structure	α-helix: negative bands at 208 and 222 nm
IR spectroscopy	vibrational frequencies	cis/trans alkenes, hydrogen bonding, conformers	C–H out-of-plane bending: trans-CH=CH ≈ 965 cm−1, cis ≈ 690 cm−1
Dipole moment measurement	molecular polarity	cis/trans isomers, conformations	cis-1,2-dichloroethene 1.9 D, trans 0 D
Electron diffraction, microwave spectroscopy	gas-phase scattering, rotational spectra	geometry of free molecules	bond lengths and angles of small molecules
Chemical methods	stereospecific reactions, chemical correlation	relative configuration	maleic acid forms a cyclic anhydride, fumaric acid does not
Molecular modelling	molecular mechanics, quantum chemistry	conformational energies, preferred geometry	butane: anti vs gauche ≈ 3.8 kJ/mol

4. Self-Control Questions (Pre-lab Preparation)

1. What is the difference between configuration and conformation? Give examples.
2. Draw the Newman projections of the conformations of butane and arrange them in order of increasing energy.
3. Why do large substituents on cyclohexane prefer the equatorial position?
4. Which method allows the absolute configuration of a molecule to be determined directly? What is its main limitation?
5. How can the cis and trans isomers of a disubstituted alkene be distinguished by 1H NMR and by IR spectroscopy?
6. Explain why cis-1,2-dichloroethene has a dipole moment while the trans isomer does not.
7. Compare the physical properties of maleic and fumaric acids and explain the differences on the basis of their structure.
8. Why does maleic acid form a cyclic anhydride on heating, while fumaric acid does not?
9. Explain the mechanism of the acid-catalysed isomerisation of maleic acid into fumaric acid.
10. What information about proteins can be obtained from circular dichroism spectra?

5. Reagents, Materials, and Equipment

Test substances: maleic acid, fumaric acid (reference sample).

Reagent	Composition / notes	Used in
Concentrated hydrochloric acid	HCl, 36%	Exp. 3
Distilled water	solubility tests, washing	Exp. 2, 3
Molecular model kits	ball-and-stick models	Exp. 1
Reference spectral data	tables of coupling constants and IR bands; spectra provided by the teacher	Exp. 4

Equipment: molecular model kits, test tubes, round-bottom flask (25–50 mL) with reflux condenser, water bath or heating mantle, boiling chips, Büchner funnel with filter flask, melting-point apparatus, balance, fume hood, safety goggles and gloves.

6. Laboratory Experiments

Experiment 1 — Molecular Models: Conformations and Configurations

Principle: ball-and-stick models help visualise conformations and configurations and establish the relationship between stereoisomers (identical, enantiomers, diastereomers).

Procedure:

1. Build a model of ethane and turn it into the staggered and eclipsed conformations; draw the Newman projections.
2. Build a model of butane and obtain the anti, gauche and eclipsed conformations by rotation about the C2–C3 bond; draw their Newman projections.
3. Build cyclohexane in the chair and boat forms and identify the axial and equatorial bonds. Place a methyl group in the axial and in the equatorial position and compare the 1,3-diaxial interactions.

4. Build both enantiomers of butan-2-ol, show that they cannot be superimposed, and assign the R/S configuration.
5. Build models of maleic and fumaric acids and show that they cannot be interconverted without breaking the π-bond.

Result: Newman and Fischer projections and conclusions for each model recorded in the protocol.

Experiment 2 — Comparison of the Physical Properties of Maleic and Fumaric Acids

Principle: cis–trans isomers are diastereomers with different physical properties. Maleic acid forms an intramolecular hydrogen bond and packs less efficiently in the crystal, so it is much more soluble in water and melts lower than fumaric acid, whose molecules form a strong intermolecular hydrogen-bonded network.

Procedure:

1. Place 0.2 g each of maleic acid and fumaric acid in two test tubes, add 1 mL of cold water to each and shake. Compare the solubility.
2. Determine the melting point of maleic acid (expected 130–135 °C). Fumaric acid does not melt in an open capillary; it sublimes on strong heating (above about 200 °C) — record this observation instead.

Result: maleic acid dissolves readily in cold water (about 78 g/100 mL), fumaric acid remains almost undissolved (about 0.6 g/100 mL); maleic acid melts at about 130–135 °C, fumaric acid sublimes without melting.

Experiment 3 — Isomerisation of Maleic Acid into Fumaric Acid

Principle: in the presence of concentrated HCl and on heating, the cis-isomer is converted into the thermodynamically more stable trans-isomer; the poorly soluble fumaric acid crystallises from the hot solution.

Procedure:

1. In a 25–50 mL round-bottom flask, dissolve 2.0 g of maleic acid in 4 mL of warm water.
2. In the fume hood, add 4 mL of concentrated HCl and 1–2 boiling chips, attach a reflux condenser and heat at gentle boiling for 15–20 minutes. Crystals of fumaric acid begin to separate from the hot solution.
3. Cool the mixture to room temperature, then in an ice bath.
4. Collect the crystals by vacuum filtration, wash them with a little cold water and dry. Weigh the product and calculate the yield.
5. Identify the product: compare its solubility in cold water with that of maleic acid (Experiment 2) and check that it does not melt below 200 °C.

$$HOOC{-}CH{=}CH{-}COOH\ (maleic\ acid,\ Z) \rightarrow HOOC{-}CH{=}CH{-}COOH\ (fumaric\ acid,\ E) \quad (HCl,\ t)$$

Result: white crystals, poorly soluble in cold water, that do not melt in an open capillary — fumaric acid. Theoretical yield: 2.0 g (isomerisation does not change the molar mass).

Experiment 4 — Assignment of Spatial Structure from Spectral Data (Desk Exercise)

Principle: vicinal 1H–1H coupling constants depend on the dihedral angle between the C–H bonds (Karplus relationship), and characteristic IR bands depend on the geometry of the molecule. These data allow the configuration of double bonds and the conformation of rings to be assigned.

Procedure:

1. Use the reference values: 3J(H,H) for trans-CH=CH 12–18 Hz, for cis-CH=CH 6–12 Hz; in cyclohexanes 3J(ax,ax) 8–13 Hz, 3J(ax,eq) and 3J(eq,eq) 2–5 Hz; IR out-of-plane C–H bending for trans-CH=CH ≈ 965 cm−1, for cis-CH=CH ≈ 690 cm−1.
2. The vinyl protons of two isomeric cinnamic acids (C6H5–CH=CH–COOH) show coupling constants of 16.0 Hz and 12.7 Hz. Assign the E and Z configurations and justify your answer.
3. Solve the additional spectral problems provided by the teacher and record your reasoning in the protocol.

Result: configurations and conformations assigned with justification recorded in the protocol.

7. Safety Precautions

- Concentrated hydrochloric acid releases corrosive HCl vapour; measure it and carry out the reflux in the fume hood.
- Maleic acid is harmful if swallowed, irritates the skin, eyes and respiratory tract and may cause skin sensitisation; avoid raising dust.
- Heat the reaction flask with a water bath or heating mantle, and add boiling chips before heating.
- Wear a lab coat, safety goggles, and gloves at all times; no food or drink in the laboratory.
- Know the location of the eyewash station, safety shower, and fire extinguisher before starting work.

8. Individual Assignment and Report Requirements

Individual assignment: (1) draw all stereoisomers of 2,3-dibromobutane and of 1,2-dimethylcyclohexane, indicate which are chiral and which are meso (achiral), and propose a method that would distinguish them. (2) Calculate the yield of fumaric acid in Experiment 3.

Protocol (laboratory report) requirements. For each experiment performed, the notebook must contain:

1. Name of the experiment and the reagents used.
2. Balanced equation(s) of the reactions.
3. Observations (colour, precipitate, gas, odour, mass, melting or boiling point) described precisely, not just “positive/negative”.
4. Calculations where required (theoretical and percent yield, specific rotation, etc.).
5. A short conclusion for each experiment linking the result to the structure of the compound.

Record the properties of the starting material, the reference sample and your product.

Property	Maleic acid (starting material)	Fumaric acid (reference)	Product of Exp. 3
Configuration (E/Z)			
Solubility in cold water			
Behaviour on heating / m.p. (°C)			
Mass (g) / yield (%)	2.0 / —	—	

9. Recommended Literature

1. Eliel, E. L., Wilen, S. H., Mander, L. N. Stereochemistry of Organic Compounds. Wiley, 1994.
2. Silverstein, R. M., Webster, F. X., Kiemle, D. J. Spectrometric Identification of Organic Compounds. Wiley.
3. Clayden, J., Greeves, N., Warren, S. Organic Chemistry. Oxford University Press.
4. Furniss, B. S. et al. (Vogel). Vogel's Textbook of Practical Organic Chemistry.
5. The official study guide and lecture notes for Chemistry of the Functional Derivatives of Organic Molecules (current program, Year 3).

Students should also consult the current national/institutional pharmacopoeia for official identification reactions and assay methods related to this topic.

Laboratory work No. 14

Diazo Coupling. Preparation of Methyl Orange

Topic: Azo coupling as electrophilic aromatic substitution. Azo dyes and acid–base indicators. Synthesis of methyl orange

1. Relevance of the Topic

Azo compounds — substances in which the –N=N– group links two aromatic rings — make up more than half of all synthetic dyes. In pharmacy and medicine they are used as acid–base indicators (methyl orange, methyl red, Congo red), as colourants and histological stains, and as drug substances: the azo dye prontosil was the first sulfonamide antibacterial drug (it is reduced in the body to sulfanilamide), and sulfasalazine is used to treat ulcerative colitis.

Azo coupling is also used in pharmaceutical analysis. Drugs containing a primary aromatic amino group (sulfonamides, procaine, benzocaine) are identified by diazotization followed by coupling with 2-naphthol to form an intensely coloured azo dye. The synthesis of methyl orange combines both steps — diazotization and coupling — in one short preparation.

2. Learning Objectives

By the end of the session, the student must know:

- The mechanism of azo coupling as electrophilic aromatic substitution in which the arenediazonium ion is a weak electrophile.
- Which aromatic compounds can act as coupling components and why; the orientation of coupling (para, or ortho if para is occupied).
- The optimal pH for coupling with phenols (weakly alkaline) and with aromatic amines (weakly acidic), and the reasons for it.
- The relationship between structure and colour in azo dyes (chromophore, auxochromes) and the mechanism of the colour change of methyl orange.

By the end of the session, the student must be able to:

- Carry out the diazotization of sulfanilic acid and its coupling with N,N-dimethylaniline at the correct temperature and pH.
- Isolate an azo dye by salting out.
- Study the indicator properties of the synthesised dye.
- Calculate the yield.

3. Theoretical Background

3.1 Diazonium salts and azo coupling

Arenediazonium salts Ar–N2+ X− are obtained by diazotization of primary aromatic amines with nitrous acid (NaNO2 + HCl) at 0–5 °C. They undergo two types of reactions: with loss of nitrogen (replacement of the diazo group by OH, Hal, CN, H) and without loss of nitrogen — azo coupling.

In azo coupling the diazonium cation acts as an electrophile. Because its positive charge is delocalised over the aromatic ring, it is a weak electrophile and reacts only with strongly activated rings — phenols

(as phenoxide ions) and aromatic amines (N,N-dialkylanilines, naphthylamines). Substitution takes place para to the activating group, or ortho if the para position is occupied.

The pH of the reaction medium is critical. Phenols couple in weakly alkaline solution (pH 8–10), where they are present as the much more reactive phenoxide ions; aromatic amines couple in weakly acidic solution (pH 4–7). In strongly acidic medium the amine is protonated and the –NHR2+ group deactivates the ring; in strongly alkaline medium the diazonium ion is converted into an unreactive diazotate Ar–N=N–O−.

3.2 Azo dyes and indicators. Methyl orange

The colour of azo dyes is due to the extended conjugated system Ar–N=N–Ar (the azo group is the chromophore). Electron-donating groups (–OH, –NR2; auxochromes) at one end of the system and electron-withdrawing groups at the other shift absorption into the visible region. Sulfonic acid groups make the dyes water-soluble.

Methyl orange is sodium 4-[4-(dimethylamino)phenylazo]benzenesulfonate. It is prepared by diazotization of sulfanilic acid and coupling of the diazonium salt with N,N-dimethylaniline. Sulfanilic acid exists as a zwitterion H3N+–C6H4–SO3− and is poorly soluble in acid, so it is first dissolved as the sodium salt in sodium carbonate solution. On acidification in the presence of nitrite, the diazonium salt separates as a fine precipitate of the inner salt −O3S–C6H4–N2+.

$$2\ H2N–C6H4–SO3H + Na2CO3 \rightarrow 2\ H2N–C6H4–SO3Na + CO2\uparrow + H2O$$

$$H2N–C6H4–SO3Na + NaNO2 + 2HCl \rightarrow -O3S–C6H4–N2+ + 2NaCl + 2H2O$$

$$-O3S–C6H4–N2+ + C6H5N(CH3)2 \rightarrow HO3S–C6H4–N=N–C6H4–N(CH3)2 \quad \textit{(red acid form)}$$

$$HO3S–C6H4–N=N–C6H4–N(CH3)2 + NaOH \rightarrow NaO3S–C6H4–N=N–C6H4–N(CH3)2 + H2O$$

(methyl orange)

As an indicator, methyl orange changes colour in the pH range 3.1–4.4. In the basic (yellow) form the molecule has the azo structure; in acidic solution it is protonated at a nitrogen atom of the azo group, giving a cation with a quinonoid structure and a different conjugated system that absorbs at a longer wavelength (red).

4. Self-Control Questions (Pre-lab Preparation)

1. Write the equation of diazotization of sulfanilic acid. Why must sulfanilic acid first be dissolved in sodium carbonate solution?
2. Why is the diazonium ion a weak electrophile? Which aromatic compounds can react with it?
3. Why is azo coupling with phenols carried out in weakly alkaline medium and with amines in weakly acidic medium? What happens if the pH is too low or too high?
4. At which position of N,N-dimethylaniline does coupling take place, and why?
5. Write the structures of the acid (red) and basic (yellow) forms of methyl orange and explain the colour change.
6. In which pH range does methyl orange change colour? For which titrations is it suitable?
7. Why is the product salted out with sodium chloride?
8. Why does reduction of an azo dye (for example, with zinc and acid) lead to decolourisation? Which products are formed from methyl orange?

9. How is the primary aromatic amino group of sulfonamides detected in pharmacopoeial analysis?
10. Explain why prontosil is inactive in vitro but active in vivo.

5. Reagents, Materials, and Equipment

Starting materials: sulfanilic acid (4-aminobenzenesulfonic acid), N,N-dimethylaniline.

Reagent	Composition / notes	Used in
Sodium carbonate	anhydrous Na2CO3	Exp. 1
Sodium nitrite	NaNO2, solid	Exp. 1
Concentrated hydrochloric acid	HCl, 36%	Exp. 1, 3
Glacial acetic acid	solvent for N,N-dimethylaniline	Exp. 1
Sodium hydroxide solution	10% aqueous NaOH	Exp. 1
Sodium chloride	solid and saturated solution — salting out	Exp. 1
Starch–iodide paper	test for excess nitrous acid	Exp. 1
Hydrochloric acid and sodium hydroxide, 0.1 M	for the indicator test (or buffer solutions pH 2, 4 and 8)	Exp. 2
Zinc dust	reducing agent	Exp. 3
Crushed ice	cooling	Exp. 1

Equipment: beakers (50–100 mL), test tubes, ice bath, thermometer, glass rods, Büchner funnel with filter flask, hot plate or water bath, indicator paper, fume hood, safety goggles and gloves.

6. Laboratory Experiments

Experiment 1 — Synthesis of Methyl Orange

Principle: diazotization of sulfanilic acid (as its sodium salt), followed by azo coupling of the diazonium salt with N,N-dimethylaniline in weakly acidic medium; the red acid form of the dye is converted into the orange sodium salt by alkali.

Procedure:

1. In a 50 mL beaker, dissolve 1.73 g (10 mmol) of sulfanilic acid and 0.53 g (5 mmol) of anhydrous Na2CO3 in 20 mL of water by warming. Cool the solution to room temperature.
2. Add a solution of 0.75 g (11 mmol) of NaNO2 in 3 mL of water.
3. Pour this solution, with stirring, into a beaker containing about 10 g of crushed ice and 2 mL of concentrated HCl. A fine white precipitate of the diazonium salt appears. Keep the mixture in an ice bath at 0–5 °C for 5–10 minutes. Check with starch–iodide paper that a slight excess of nitrous acid is present (immediate blue colour).
4. In a test tube, in the fume hood, mix 1.3 mL (1.21 g, 10 mmol) of N,N-dimethylaniline with 1 mL of glacial acetic acid.
5. Add the N,N-dimethylaniline solution to the cold suspension of the diazonium salt with vigorous stirring. A red colour develops (the acid form of the dye). Keep the mixture in the ice bath for 10 minutes, stirring from time to time.
6. Slowly add 12–15 mL of 10% NaOH with stirring until the mixture is alkaline (indicator paper): the colour changes to orange and the sodium salt of methyl orange begins to separate.

7. Heat the mixture to boiling to dissolve most of the dye, add about 5 g of NaCl, stir until it dissolves, and allow the mixture to cool to room temperature and then in an ice bath.
8. Collect the orange crystals by vacuum filtration, wash them with a small amount of saturated NaCl solution and dry. (The dye can be recrystallised from a small volume of hot water.) Weigh the product and calculate the yield.

Result: orange-yellow crystalline leaflets of methyl orange, C14H14N3NaO3S, M = 327.33 g/mol; it does not melt but decomposes above 300 °C. Theoretical yield: 3.27 g.

Experiment 2 — Indicator Properties of Methyl Orange

Principle: the dye exists in two forms with different conjugated systems: red (protonated, quinonoid) below pH 3.1 and yellow (azo form) above pH 4.4.

Procedure:

1. Dissolve a few crystals of the synthesised dye in 10 mL of water.
2. Place 2 mL of this solution in each of three test tubes. Add 1 mL of 0.1 M HCl to the first, 1 mL of water to the second and 1 mL of 0.1 M NaOH to the third. Compare the colours.
3. Add 0.1 M NaOH dropwise to the first tube and observe the colour change.

$$\textit{yellow azo form } (pH > 4.4) + H+ \rightleftharpoons \textit{red quinonoid form } (pH < 3.1)$$

Result: red in acid, yellow in alkali, orange in the transition range; the change is reversible.

Experiment 3 — Reductive Cleavage of the Azo Group

Principle: strong reducing agents cleave the –N=N– bond and destroy the chromophore; methyl orange is reduced to sulfanilic acid and N,N-dimethyl-p-phenylenediamine, and the solution becomes colourless.

Procedure:

1. To 2 mL of methyl orange solution add 1 mL of concentrated HCl (red colour) and a small spatula-tip of zinc dust.
2. Warm the tube gently in a water bath in the fume hood (hydrogen is evolved) and observe.

$$NaO3S–C6H4–N=N–C6H4–N(CH3)2 + 4[H] \rightarrow NaO3S–C6H4–NH2 + H2N–C6H4–N(CH3)2$$

Result: the red colour gradually fades and the solution becomes colourless.

Practical significance: reductive cleavage of the azo bond is a metabolic pathway of azo drugs: prontosil is reduced by azoreductases to the active sulfanilamide, and sulfasalazine is cleaved by gut bacteria to 5-aminosalicylic acid and sulfapyridine.

7. Safety Precautions

- N,N-Dimethylaniline is toxic by inhalation, ingestion and skin contact (it causes methaemoglobinaemia) and is suspected of causing cancer. Measure it in the fume hood with gloves and avoid any skin contact.
- Sodium nitrite is toxic if swallowed and is an oxidiser. Diazonium salts must be kept cold and used immediately in solution; never isolate or dry them (dry diazonium salts can explode).
- Concentrated HCl and glacial acetic acid are corrosive; NaOH solution is caustic.
- Zinc dust with acid releases flammable hydrogen — no open flames nearby.
- Methyl orange stains skin and clothing.

- Wear a lab coat, safety goggles, and gloves at all times; no food or drink in the laboratory.
- Know the location of the eyewash station, safety shower, and fire extinguisher before starting work.
- Destroy any unused diazonium suspension by warming it in the fume hood until nitrogen evolution stops, and collect all residues in the designated containers.

8. Individual Assignment and Report Requirements

Individual assignment: (1) calculate the theoretical and percent yield of methyl orange. (2) Write the equations for the synthesis of methyl red (from anthranilic acid and N,N-dimethylaniline) and of the orange dye obtained from benzenediazonium chloride and 2-naphthol; state the pH at which each coupling should be carried out.

Protocol (laboratory report) requirements. For each experiment performed, the notebook must contain:

1. Name of the experiment and the reagents used.
2. Balanced equation(s) of the reactions.
3. Observations (colour, precipitate, gas, odour, mass, melting or boiling point) described precisely, not just “positive/negative”.
4. Calculations where required (theoretical and percent yield, specific rotation, etc.).
5. A short conclusion for each experiment linking the result to the structure of the compound.

Record your results in the table below.

Stage / test	Conditions	Observation	Conclusion
Dissolution of sulfanilic acid	Na2CO3, warming		
Diazotization	NaNO2, HCl, 0–5 °C		
Coupling	N,N-dimethylaniline, CH3COOH		
Addition of alkali	NaOH		
Indicator test	HCl / H2O / NaOH		
Reduction	Zn, HCl		
Product	mass ____ g	yield ____ %	

9. Recommended Literature

1. Clayden, J., Greeves, N., Warren, S. Organic Chemistry. Oxford University Press.
2. Morrison, R. T., Boyd, R. N. Organic Chemistry.
3. Furniss, B. S. et al. (Vogel). Vogel's Textbook of Practical Organic Chemistry.
4. Zollinger, H. Color Chemistry: Syntheses, Properties, and Applications of Organic Dyes and Pigments. Wiley-VCH.
5. The official study guide and lecture notes for Chemistry of the Functional Derivatives of Organic Molecules (current program, Year 3).

Students should also consult the current national/institutional pharmacopoeia for official identification reactions and assay methods related to this topic.

Laboratory work No. 15

Carbohydrates. Monosaccharides. Qualitative Reactions

Topic: Structure and properties of monosaccharides. Qualitative reactions of aldoses, ketoses and disaccharides

1. Relevance of the Topic

Carbohydrates are the main source of energy for living organisms and the structural basis of nucleic acids, glycoproteins and glycolipids. In pharmacy, glucose solutions are used for infusion and parenteral nutrition, fructose, lactose and sucrose serve as excipients and sweeteners, and many important drugs are glycosides (cardiac glycosides, aminoglycoside antibiotics).

Qualitative reactions of monosaccharides are used to identify drug substances and in clinical laboratory diagnostics. The reduction of copper(II) and silver(I) compounds by glucose was the historical basis for detecting glucose in urine in diabetes mellitus, and Seliwanoff's test distinguishes fructose from glucose.

2. Learning Objectives

By the end of the session, the student must know:

- The classification of monosaccharides (aldoses and ketoses; trioses, pentoses, hexoses) and the D/L series.
- The open-chain (Fischer) and cyclic (Haworth) forms of glucose and fructose; pyranoses and furanoses; α- and β-anomers; mutarotation.
- The origin of the reducing properties of monosaccharides and the interconversion of aldoses and ketoses in alkaline medium.
- The chemistry of the Molisch, Trommer, Fehling, Tollens, Barfoed and Seliwanoff tests.

By the end of the session, the student must be able to:

- Detect carbohydrates in a sample and distinguish reducing from non-reducing sugars.
- Distinguish an aldose from a ketose and a monosaccharide from a disaccharide.
- Identify glucose, fructose, sucrose and maltose (lactose) in unlabelled solutions.

3. Theoretical Background

3.1 Structure of monosaccharides

Monosaccharides are polyhydroxy aldehydes (aldoses) or polyhydroxy ketones (ketoses). The most important are the aldohexose D-glucose and the ketohexose D-fructose. The D/L series is determined by the configuration of the stereogenic centre farthest from the carbonyl group (C5 in hexoses), by comparison with D-glyceraldehyde.

In solution, monosaccharides exist mainly as cyclic hemiacetals formed by intramolecular nucleophilic addition of a hydroxyl group to the carbonyl group: six-membered pyranoses (glucose) or five-membered furanoses (the fructose unit in sucrose). Cyclisation creates a new stereogenic centre — the anomeric carbon — giving α- and β-anomers. In solution the anomers interconvert through the open-chain form (mutarotation): the specific rotation of freshly dissolved α-D-glucopyranose (+112°) or

β-D-glucopyranose (+19°) changes to the equilibrium value of +52.7° (about 36% α and 64% β; less than 0.1% open-chain form).

3.2 Chemical basis of the qualitative reactions

Because the cyclic forms are in equilibrium with the open-chain form, monosaccharides show the properties of both polyhydric alcohols and carbonyl compounds.

- **Polyol properties:** vicinal hydroxyl groups dissolve Cu(OH)2 in alkaline medium at room temperature to form a deep-blue chelate complex (first stage of Trommer's test).
- **Reducing properties:** the free (or potentially free) carbonyl group reduces Cu(II) to red Cu2O (Trommer, Fehling and Benedict tests) and Ag(I) to metallic silver (Tollens' test), and the aldose is oxidised to an aldonic acid. Sugars with a free hemiacetal hydroxyl (glucose, fructose, maltose, lactose) are reducing; sucrose, in which both anomeric carbons are bound in the glycosidic bond, is non-reducing.
- **Ketoses also reduce:** in alkaline medium fructose isomerises through an enediol intermediate into glucose and mannose (Lobry de Bruyn–van Ekenstein transformation), so it also gives positive Fehling and Tollens tests.
- **Barfoed's test:** copper(II) acetate in dilute acetic acid is a weak oxidant; in this slightly acidic medium monosaccharides reduce it within 2–3 minutes, while reducing disaccharides react much more slowly.
- **Dehydration reactions:** concentrated acids dehydrate pentoses to furfural and hexoses to 5-hydroxymethylfurfural, which condense with phenols to coloured products. Molisch's test (α-naphthol, concentrated H2SO4) is general for all carbohydrates; in Seliwanoff's test (resorcinol, HCl) ketoses are dehydrated much faster than aldoses and give a cherry-red colour within 1–2 minutes.

$$CH2OH–(CHOH)4–CHO + 2Cu(OH)2 \rightarrow CH2OH–(CHOH)4–COOH + Cu2O\downarrow + 2H2O$$

$$CH2OH–(CHOH)4–CHO + 2[Ag(NH3)2]OH \rightarrow CH2OH–(CHOH)4–COONH4 + 2Ag\downarrow + 3NH3 + H2O$$

4. Self-Control Questions (Pre-lab Preparation)

1. Write the Fischer projections of D-glucose and D-fructose and their Haworth formulas (α-D-glucopyranose, β-D-fructofuranose).
2. What is mutarotation? Explain it using D-glucose as an example.
3. Which property of monosaccharides is detected by the reaction with Cu(OH)2 at room temperature, and which on heating?
4. Why is fructose, a ketose, a reducing sugar?
5. Why is sucrose non-reducing, while maltose and lactose are reducing?
6. What is the principle of Barfoed's test? What does it allow you to distinguish?
7. Explain the chemistry of Seliwanoff's test. Why do ketoses react faster than aldoses?
8. Which carbohydrates give a positive Molisch test?
9. Propose a scheme to identify glucose, fructose, sucrose and maltose in four unlabelled test tubes.
10. What is the clinical significance of detecting glucose in urine?

5. Reagents, Materials, and Equipment

Test substances: 1–5% aqueous solutions of glucose, fructose, sucrose and maltose (or lactose).

Reagent	Composition / notes	Used in
Molisch reagent	5% α-naphthol in ethanol (freshly prepared); concentrated H2SO4	Exp. 1
Copper(II) sulfate / sodium hydroxide	5% CuSO4; 10% NaOH	Exp. 2
Fehling's solutions A and B (or Benedict's reagent)	A: ≈7% CuSO4·5H2O; B: potassium sodium tartrate + NaOH	Exp. 3
Tollens' reagent	2% AgNO3 + 10% NaOH + dilute NH3 — prepared immediately before use	Exp. 4
Barfoed's reagent	copper(II) acetate in dilute acetic acid	Exp. 5
Seliwanoff's reagent	0.05% resorcinol in hydrochloric acid (conc. HCl : water = 1 : 2)	Exp. 6

Equipment: test tubes and rack, pipettes, boiling water bath, water bath at 50–60 °C, stopwatch, fume hood, safety goggles and gloves.

6. Laboratory Experiments

Experiment 1 — Molisch's Test (General Test for Carbohydrates)

Principle: concentrated sulfuric acid dehydrates carbohydrates (after hydrolysis of glycosidic bonds) to furfural derivatives, which condense with α-naphthol to form a violet product.

Procedure:

1. To 1 mL of each sugar solution add 2–3 drops of Molisch reagent and mix.
2. Tilt the test tube and carefully pour 1 mL of concentrated H2SO4 down the wall so that it forms a separate lower layer. Do not shake.

hexose → 5-hydroxymethylfurfural (conc. H2SO4, −3H2O); + α-naphthol → violet condensation product

Result: a violet ring at the interface of the two layers with all the carbohydrates tested.

Experiment 2 — Trommer's Test

Principle: at room temperature, Cu(OH)2 is dissolved by the polyol structure (chelate formation — a test for vicinal diols); on heating, reducing sugars reduce Cu(II) to Cu(I).

Procedure:

1. To 1 mL of each sugar solution add 1 mL of 10% NaOH and 3–5 drops of 5% CuSO4. Shake and note the colour at room temperature.
2. Place the tubes in a boiling water bath for 2–3 minutes and observe the colour change.

sugar (polyol) + Cu(OH)2 → deep-blue copper(II) chelate

CH2OH–(CHOH)4–CHO + 2Cu(OH)2 → CH2OH–(CHOH)4–COOH + Cu2O↓ + 2H2O

Result: at room temperature all sugars give a clear deep-blue solution (positive for polyols, including sucrose); on heating, glucose, fructose and maltose give first a yellow (CuOH) and then a red (Cu2O) precipitate; sucrose stays blue.

Experiment 3 — Fehling's (or Benedict's) Test

Principle: reducing sugars reduce the deep-blue copper(II) tartrate (or citrate) complex to red copper(I) oxide on heating.

Procedure: mix 1 mL of Fehling's solution A and 1 mL of Fehling's solution B (or take 2 mL of Benedict's reagent), add 1 mL of the sugar solution and heat in a boiling water bath for 2–3 minutes.

$$RCHO + 2Cu2+ + 5OH- \rightarrow RCOO- + Cu2O\downarrow + 3H2O$$

Result: a brick-red precipitate with glucose, fructose and maltose (lactose); sucrose — negative (the solution remains blue).

Experiment 4 — Tollens' Test (Silver Mirror)

Principle: reducing sugars reduce the diamminesilver(I) complex to metallic silver.

Procedure: to 2 mL of freshly prepared Tollens' reagent in a clean test tube add 3–5 drops of the sugar solution and warm in a water bath at 50–60 °C for a few minutes without shaking.

$$CH2OH-(CHOH)4-CHO + 2[Ag(NH3)2]OH \rightarrow CH2OH-(CHOH)4-COONH4 + 2Ag\downarrow + 3NH3 + H2O$$

Result: a silver mirror (or grey-black precipitate) with glucose, fructose and maltose; sucrose — negative.

Important: prepare Tollens' reagent immediately before use and never store it; acidify and discard the residues right after the experiment.

Experiment 5 — Barfoed's Test (Monosaccharides vs Disaccharides)

Principle: in weakly acidic medium, copper(II) acetate is a weak oxidant: monosaccharides reduce it quickly, reducing disaccharides only after prolonged heating.

Procedure: to 2 mL of Barfoed's reagent add 1 mL of the sugar solution (glucose, fructose, maltose or lactose). Place all the tubes in a boiling water bath at the same time and note when a red precipitate appears; stop the observation after 5 minutes.

Result: monosaccharides — a red precipitate of Cu2O within 2–3 minutes; disaccharides — no precipitate within this time (a precipitate may appear after 10 minutes or more because of partial hydrolysis, and must not be interpreted as positive).

Experiment 6 — Seliwanoff's Test (Ketoses vs Aldoses)

Principle: in hot hydrochloric acid, ketohexoses are rapidly dehydrated to 5-hydroxymethylfurfural, which condenses with resorcinol to form a cherry-red product; aldohexoses react much more slowly.

Procedure: to 2 mL of Seliwanoff's reagent add 0.5 mL of the sugar solution (glucose, fructose, sucrose). Place the tubes in a boiling water bath at the same time and observe for no more than 1–2 minutes.

$$fructose \rightarrow 5\text{-}hydroxymethylfurfural\ (HCl,\ t,\ -3H2O);\ + resorcinol \rightarrow cherry\text{-}red\ product$$

Result: fructose — cherry-red colour within about 1 minute; sucrose — red colour after a short delay (acid hydrolysis releases fructose); glucose — no colour, or only faint pink after prolonged heating.

7. Safety Precautions

- Concentrated sulfuric acid (Molisch test) and hydrochloric acid (Seliwanoff's reagent) are corrosive. Add H2SO4 slowly down the wall of the tube and never point a heated tube at anyone.

- Tollens' reagent must be freshly prepared and never stored: on standing it can deposit explosive silver compounds. Acidify and discard residues immediately.
- α-Naphthol and resorcinol are harmful and irritating; NaOH solutions are caustic.
- Wear a lab coat, safety goggles, and gloves at all times; no food or drink in the laboratory.
- Know the location of the eyewash station, safety shower, and fire extinguisher before starting work.
- Collect silver- and copper-containing residues in the designated heavy-metal waste container.

8. Individual Assignment and Report Requirements

Individual assignment: four unlabelled test tubes contain solutions of glucose, fructose, sucrose and maltose. Propose the shortest sequence of tests from this work that identifies each solution, and justify each step with an equation or a structural argument.

Protocol (laboratory report) requirements. For each experiment performed, the notebook must contain:

1. Name of the experiment and the reagents used.
2. Balanced equation(s) of the reactions.
3. Observations (colour, precipitate, gas, odour, mass, melting or boiling point) described precisely, not just “positive/negative”.
4. Calculations where required (theoretical and percent yield, specific rotation, etc.).
5. A short conclusion for each experiment linking the result to the structure of the compound.

Mark each result as “+” or “–” and note the colour or the time of appearance.

Test	Glucose	Fructose	Sucrose	Maltose / lactose	Detected feature
Molisch					carbohydrate
Trommer, cold					polyol (vicinal OH)
Trommer, heating					reducing sugar
Fehling / Benedict					reducing sugar
Tollens					reducing sugar
Barfoed					monosaccharide
Seliwanoff					ketose

9. Recommended Literature

1. Clayden, J., Greeves, N., Warren, S. Organic Chemistry. Oxford University Press.
2. Morrison, R. T., Boyd, R. N. Organic Chemistry.
3. Nelson, D. L., Cox, M. M. Lehninger Principles of Biochemistry.
4. Plummer, D. T. An Introduction to Practical Biochemistry.
5. Furniss, B. S. et al. (Vogel). Vogel's Textbook of Practical Organic Chemistry.
6. The official study guide and lecture notes for Chemistry of the Functional Derivatives of Organic Molecules (current program, Year 3).

Students should also consult the current national/institutional pharmacopoeia for official identification reactions and assay methods related to this topic.